Canonical basis of the quantized enveloping algebra of finite type

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Abstract

This note is devoted to the construction of the canonical basis of the quantized enveloping algebra $\mathbf{U}$ associated to a symmetric Cartan matrix $C = (a_{ij})$. Following Lusztig's original paper [Lus90a], we establish a PBW type basis of $\mathbf{U}$ via Lusztig's symmetries T_i, and use them to construct the canonical basis $\mathbf{B}$ by solving a system of semilinear equations. We then show that the transition coefficients of $\mathbf{B}$ with respect to the PBW type basis we constructed can be related to the intersection cohomology of certain orbits in the representation space of the corresponding quiver, which implies the positivity of the canonical basis. Finally, we present a purity result for the intersection cohomology of these orbits given in [Lus90a], which is analogous to the purity of intersection cohomology of Schubert varieties.

Contents

1 Notations

Let $(a_{ij})_{1\le i,j\le n}$ be a symmetric positive Cartan matrix, and write $\mathbb{I}=[1,n]$. Let $\mathbb{Z}^{\mathbb{I}}$ be the free abelian group with basis $\{\alpha_1,\dots,\alpha_n\}$ and define an inner product on $\mathbb{Z}^{\mathbb{I}}$ by $(\alpha_i,\alpha_j)=a_{ij}$. Let $R=\{\alpha\in\mathbb{Z}^{\mathbb{I}}\mid(\alpha,\alpha)=2\}$ be the corresponding root system, whose set of simple roots is $\{\alpha_1,\dots,\alpha_n\}$. Let R^+ and $R^-=-R^+$ be the corresponding sets of positive and negative roots, respectively. Any $\alpha\in R$ defines a reflection $s_\alpha:\mathbb{Z}^{\mathbb{I}}\to\mathbb{Z}^{\mathbb{I}}$, $z\mapsto z-(z,\alpha)\alpha$. We write s_i instead of s_{α_i}. Let W be the Weyl group of R, which is generated by reflections s_i ($1\le i\le n$). Let $\ell(w)$ be the usual length function on W with respect to the generators $\{s_1,\dots,s_n\}$ and w_0 be the unique element of W of maximal length.

Let $\mathbf{U}$ be the Drinfeld-Jimbo quantized enveloping algebra corresponding to (a_{ij}). With the notation of [Lus90b], this is the $\mathbb{Q}(v)$-algebra with generators E_i, F_i, $K_i^{\pm1}$ ($i\in\mathbb{I}$) and relations

$$\begin{gathered}
K_iK_j=K_jK_i,\quad K_iK_i^{-1}=1,\\
K_iE_j=v^{a_{ij}}E_jK_i,\quad K_iF_j=v^{-a_{ij}}F_jK_i,\\
[E_i,E_j]=\delta_{ij}\frac{K_i-K_i^{-1}}{v-v^{-1}},\\
\sum_{s=0}^{1-a_{ij}}(-1)^sE_i^{(s)}E_jE_i^{(1-a_{ij}-s)}=0,\\
\sum_{s=0}^{1-a_{ij}}(-1)^sF_i^{(s)}F_jF_i^{(1-a_{ij}-s)}=0,
\end{gathered}$$

where $E_i^{(n)}=E_i^n/[n]!$, $F_i^{(n)}=F_i^n/[n]!$. Let $\mathbf{U}^+$ be the $\mathbb{Q}(v)$-subalgebra generated by the elements E_i and $\mathbf{U}^-$ be that generated by F_i. Let $\mathbf{U}^0$ be the $\mathbb{Q}(v)$-subalgebra of $\mathbf{U}$ generated by the elements $K_i^{\pm1}$. Let U^+ (resp. U^-) be the A-subalgebra of $\mathbf{U}^+$ (resp. $\mathbf{U}^-$) generated by the elements $E_i^{(N)}$, $F_i^{(N)}$, respectively, where $A=\mathbb{Z}[v,v^{-1}]$. Let $^-:\mathbf{U}\to\mathbf{U}$ be the $\mathbb{Q}$-algebra involution defined by

$$E_i\mapsto E_i,\quad F_i\mapsto F_i,\quad K_i\mapsto K_i^{-1}$$

for all i and $v\mapsto v^{-1}$. For each $\mu\in\mathbb{N}^{\mathbb{I}}$, we denote by $\mathbf{U}_\mu^+$ the set of elements of degree μ, where we set $\deg(K_i^{\pm1})=0$, $\deg(E_i)=\alpha_i$, $\deg(F_i)=-\alpha_i$. The subspaces $\mathbf{U}_\mu^+$ form a direct sum decomposition of $\mathbf{U}^+$, and there intersection with U^+ form a direct sum decomposition of U^+. We also need the $\mathbb{Q}(v)$-algebra anti-automorphism τ of $\mathbf{U}$, which is defined by

$$E_i\mapsto E_i,\quad F_i\to F_i,\quad K_i\mapsto K_i^{-1}.$$

2 Lusztig's symmetries

In this section we briefly review the definition of Lusztig's symmetries T_w on the algebra $\mathbf{U}$, which will be used to define a PBW type basis of $\mathbf{U}$. First, for $i\in\mathbb{I}$, Lusztig defined a $\mathbb{Q}(v)$-algebra automorphism $T_i:\mathbf{U}\to\mathbf{U}$ given by the formulas[1]

$$\begin{gathered}
T_i(E_i)=-K_i^{-1}F_i,\quad T_i(F_i)=-E_iK_i,\quad T_i(K_j)=K_{s_i(j)},\\
T_i(E_j)=\begin{cases}E_j & \text{if } a_{ij}=0\\ E_jE_i-v^{-1}E_iE_j & \text{if } a_{ij}=-1\end{cases},\\
T_i(F_j)=\begin{cases}F_j & \text{if } a_{ij}=0\\ F_iF_j-vF_jF_i & \text{if } a_{ij}=-1\end{cases},
\end{gathered}$$

[1] We follow the notations of [Lus90a], which is slightly different from [Jan96].

The family T_i are called *Lusztig's symmetries*, and they satisfy *braid group relations* (cf. [Jan96, Chapter 8]):

$$\begin{cases} T_iT_j = T_jT_i & \text{if } a_{ij} = 0, \\ T_iT_jT_i = T_jT_iT_j & \text{if } a_{ij} = -1. \end{cases}$$

We also note that if $i \neq j$ are such that $a_{ij} = -1$, then the above formulas imply

$$T_i(E_j) = E_jE_i - v^{-1}E_iE_j = T_j^{-1}(E_i), \tag{2.1}$$

$$T_i^{-1}(E_j) = E_iE_j - v^{-1}E_jE_i = T_j(E_i), \tag{2.2}$$

$$T_i(F_j) = F_iF_j - vF_jF_i = T_j^{-1}(F_i), \tag{2.3}$$

$$T_i^{-1}(F_j) = F_jF_i - vF_iF_j = T_j(F_i), \tag{2.4}$$

and in particular we have

$$\begin{aligned} T_iT_j(E_i) = E_j, \quad T_jT_i(E_j) = E_i, \\ T_iT_j(F_i) = F_j, \quad T_jT_i(F_j) = F_i. \end{aligned} \tag{2.5}$$

Now recall that the Weyl group W is generated by reflections s_i for $i \in \mathbb{I}$. If $w \in W$ is written as $w = s_{i_1}s_{i_2}\cdots s_{i_t}$ with t minimal, then this is called a reduced expression of w, and t is called the length $\ell(w)$ of w. For distinct simple roots α_i and α_i, if s_is_j has order 2 (i.e. $s_is_j = s_js_i$), and w has a reduced expression of the form

$$w = s_{i_1}\cdots s_{i_{r-1}}s_is_js_{i_{r+2}}\cdots s_{i_t}$$

then

$$w = s_{i_1}\cdots s_{i_{r-1}}s_js_is_{i_{r+2}}\cdots s_{i_t}$$

is also a reduced expression of w. If s_is_j has order 3 (i.e. $s_is_js_i = s_js_is_j$), and if w has a reduced expression of the form

$$w = s_{i_1}\cdots s_{i_{r-1}}s_is_js_is_{i_{r+3}}\cdots s_{i_t}$$

then

$$w = s_{i_1}\cdots s_{i_{r-1}}s_js_is_js_{i_{r+3}}\cdots s_{i_t}$$

is also a reduced expression of w. We call the transition from one reduced expression to another one an *elementary move*, if it has one of the forms described above. It turns out that given two expressions of an element $w \in W$, one can get from one to the other by a sequence of elementary moves, cf. [Bou02, Ch. IV, §1, exerc. 13].

This result implies that we can define for each $w \in W$ an automorphism T_w on $\mathbf{U}$ as follows. For $w = 1$ set $T_w = 1$ (the identity). For $w \neq 1$ choose a reduced expression $w = s_{i_1}\cdots s_{i_t}$ and set

$$T_w = T_{i_1}\cdots T_{i_t}.$$

It can be shown that this definition is independent of the choice of reduced expressions, and it is clear that we have

$$T_w^{-1} = \tau \circ T_{w^{-1}} \circ \tau.$$

Lemma 2.1. *Let $i \neq j$ and $w \in W$ be in the subgroup generated by s_i and s_j. If $w(\alpha_i) > 0$, then $T_w(E_i)$ is contained in the subalgebra of $\mathbf{U}$ generated by E_i and E_j. If $w(\alpha_i) = \alpha_k$, then $T_w(E_i) = E_k$.*

Proof. The claim is trivial for $w = 1$, where we have $T_w(E_i) = E_i$, so assume that $w \neq 1$. Let m be the order of s_is_j. If $m = 2$, then s_i commutes with s_j, so we must have $w = s_j$ and $T_w(E_i) = E_i$. If $m = 3$, then $w \in \{s_j, s_is_j\}$ and thus $T_w(E_i)$ is in the subalgebra generated by $E_iE_j - v^{-1}E_jE_i$ and E_j, in view of (2.2) and (2.5). This proves the lemma. □

Proposition 2.2. *Let $w \in W$ and $i \in \mathbb{I}$. If $w(\alpha_i) > 0$, then $T_w(E_i) \in \mathbf{U}^+$. If $w(\alpha_i) = \alpha_k$, then $T_w(E_i) = E_k$.*

Proof. We use induction on $\ell(w)$. If $\ell(w) = 0$, then $w = 1$ and $T_w = 1$, so the claim is obvious in that case. Suppose now that $\ell(w) > 0$, then there is $j \in \mathbb{I}$ with $w(\alpha_j) < 0$. We have obviously $j \neq i$. There is now a decomposition $w = w'w''$ with w'' in the subgroup generated by s_j and s_i such that $w'(\alpha_j) > 0$ and $w'(\alpha_i) > 0$; we have then $\ell(w) = \ell(w') + \ell(w'')$. Now $w(\alpha_i) > 0$ and $w(\alpha_j) < 0$ implies that $w''(\alpha_i) > 0$ and $w''(\alpha_j) < 0$ (both $w''(\alpha_i)$ and $w''(\alpha_j)$ are contained in $R \cap (\mathbb{Z}\alpha_i + \mathbb{Z}\alpha_j)$, and w' maps positive resp. negative roots in this intersection to positive resp. negative roots in R). This implies in particular that $w'' \neq 1$ and thus $\ell(w') < \ell(w)$, so we can apply induction to w' and get $T_{w'}(E_i) \in \mathbf{U}^+$ and $T_{w'}(E_j) \in \mathbf{U}^+$. Now Lemma 2.1 implies that $T_{w''}(E_i)$ is contained in the subalgebra of $\mathbf{U}$ generated by E_i and E_j, so the first claim follows since $T_w = T_{w'}T_{w''}$ in this case.

In order to get the claim in the case $w(\alpha_i) = \alpha_k$ is a simple root, we have to show that $w''(\alpha)$ is a simple root in that case. Then induction applied to w' and $w''(\alpha_i)$ (together with the last part of Lemma 2.1) will yield the claim. Now if $w''(\alpha_i)$ is not a simple root, then $w''(\alpha_i) > 0$ and $w''(\alpha_i) \in R \cap (\mathbb{Z}\alpha_i + \mathbb{Z}\alpha_j)$ imply that $w''(\alpha_i) = a\alpha_i + b\alpha_j$ with $a, b > 0$. Then $w(\alpha_i) = w'w''(\alpha_i) = aw'(\alpha_i) + bw'(\alpha_j)$ cannot be simple since $w'(\alpha_i) > 0$ and $w'(\alpha_j) > 0$, so $w''(\alpha_i)$ is simple and the second claim follows. □

3 A PBW type basis of U

We now construct a PBW type basis of the algebra $\mathbf{U}$ using a reduced expression of the longest element w_0 of the Weyl group W. It turns out that every reduced expression $\mathbf{i}$ gives a basis $B_{\mathbf{i}}$. However, the $\mathbb{Z}[v^{-1}]$-lattice $\mathscr{L}$ generated by $B_{\mathbf{i}}$ is independent of the choice of $\mathbf{i}$, and our canonical basis can be extracted from $\mathscr{L}$.

To begin with, we consider a reduced expression $w = s_{i_1} \cdots s_{i_t}$. For nonnegative integers $c_1, \ldots, c_t$, we consider the product

$$E_{i_1}^{(c_1)} T_{i_1}(E_{i_2}^{(c_2)}) \cdots T_{i_1} \cdots T_{i_{t-1}}(E_{i_t}^{(c_t)}). \tag{3.1}$$

Note that for each $k \leq \nu$, $s_{i_1}s_{i_2} \cdots s_{i_k}$ is a reduced expression, so we have

$$T_{s_{i_1}s_{i_2}\cdots s_{i_k}} = T_{s_{i_1}} T_{s_{i_2}} \cdots T_{s_{i_k}}.$$

Now Proposition 2.2 implies that for all $k < \nu$,

$$T_{i_1}T_{i_2} \cdots T_{i_k}(E_{i_{k+1}}) \in \mathbf{U}^+$$

so the elements in (3.1) are in $\mathbf{U}^+$.

We now prove the linearly independence of the elements in (3.1). This follows from the following general lemma:

Lemma 3.1. *Let $i \in \mathbb{I}$ and $u_0, u_1, \ldots, u_m \in \mathbf{U}^+$ with $T_i^{-1}(u_k) \in \mathbf{U}^+$ for all k. If $\sum_{k=0}^m u_k E_i^{(k)} = 0$ or $\sum_{k=0}^m E_i^{(k)} u_k = 0$, then $u_k = 0$ for all k.*

Proof. Suppose that $\sum_{k=0}^m u_k E_i^k = 0$. If we apply T_i^{-1}, then

$$0 = T_i^{-1}\Big(\sum_{k=0}^m u_k E_i^k\Big) = \sum_{k=0}^m T_i^{-1}(u_k) T_i^{-1}(E_i)^k = T_i^{-1}(-K_i^{-1}F_i)^k.$$

Since the multiplication map $\mathbf{U}^+ \otimes \mathbf{U}^{\leq 0} \to \mathbf{U}$ is an isomorphism of vector spaces, and since the $(-K_i^{-1}F_i)^k$ are linearly independent vectors in $\mathbf{U}^{\leq 0}$ while all $T_i^{-1}(u_k)$ are in $\mathbf{U}^+$, we get $T_i^{-1}(u_k) = 0$ for all k, hence $k = 0$. The case for $\sum_{k=0}^m E_i^{(k)} u_k = 0$ can be proved similarly. □

Using Lemma 3.1, we can in fact prove that for any reduced expression $w = s_{i_1} \cdots s_{i_t}$ for $w \in W$, the products

$$E_{i_1}^{(c_1)} T_{i_1}(E_{i_2}^{(c_2)}) \cdots T_{i_1} T_{i_2} \cdots T_{i_{t-1}}(E_{i_t}^{(c_t)}) \tag{3.2}$$

are linearly independent (for all sequences $c_1, \dots, c_t$ of nonnegative intergers). We do this by induction on t. For $t = 0$, we have in (3.2) only the empty product,which is equal to 1 and linearly independent. Now suppose that $t > 0$. Then the products in (3.2) have the form $E_i^{(k)} T_i(v_j)$ where $i = i_1$ and where the v_j are the products as in (3.2) constructed for the reduced expression $s_{i_2} \cdots s_{i_t}$ of $s_i w$. If we have scalars a_{kj} with $\sum_{k,j} a_{kj} E_i^{(k)} T_i(v_j)$, then we can apply (a) with $u_k = T_i(a_{kj} v_j)$ and get $\sum_j a_{kj} v_j = 0$ for all k. By our induction on t the v_j are linearly independent, so we get indeed $a_{kj} = 0$ for all k and j.

The fact that $E_{\mathbf{i}}^{\mathbf{c}}$ generate $\mathbf{U}^+$ requires more work. For its proof, it is convenient to consider any reduced expression $w = s_{i_1} \cdots s_{i_t}$ of an element in $w \in W$, rather than that of w_0. Again we can introduce the products (3.2), and the following proposition is true:

Proposition 3.2. (a) *The subspace spanned by all products as in (3.2) depends only on w, not on the chosen reduced expression.*

(b) *Let $i \neq j$. If w is the longest element in the subgroup of W generated by s_i and s_j, then the products in (3.2) span the subalgebra of $\mathbf{U}$ generated by E_i and E_j.*

Proof. We first show that (b) implies (a) by induction on $\ell(w)$. For $\ell(w) < 1$ there is only one reduced expression, so the claim holds. Suppose now that $\ell(w) > 1$ and that the claim holds for all w' with $\ell(w') < \ell(w)$. Denote by $\mathbf{U}^+[w']$ the space spanned by the products as in (3.2) for w' instead of w. Let $w = s_{j_1} s_{j_2} \cdots s_{j_t}$ be another reduced expression. We may assume that we get it from the first one by an elementary move. Set $i = i_1$ and $j = j_1$.

In case $i = j$, set $w' = s_i w$. Then the subspace spanned by the elements as in (3.2) is (for both expressions) equal to

$$\Big(\sum_{k \geq 0} \Bbbk E_i^{(k)} \Big) \cdot T_i(\mathbf{U}^+[w']).$$

This follows from our induction applied to w': The two reduced expressions lead just to two systems of generators for the subspace $\mathbf{U}^+[w']$.

Now suppose that $i \neq j$. Then the elementary move from one reduced expression to the other changes the first factor of the reduced expression, hence has to take place at the beginning of the expression. Let w'' be longest element in the subgroup of W generated by s_i and s_j. Then our two reduced expressions for w have to begin with the two reduced expressions of w'' and to coincide after that, so we have a decomposition $w = w'' w'$ with $\ell(w) = \ell(w'') + \ell(w')$. Using (b) we see that the span of the products in (3.2) is equal to $T_{w''}(\mathbf{U}^+[w'])\mathbf{U}^+[w'']$ for both expressions.

We now turn to the proof of (b). Let $i, j \in \mathbb{I}$ with $i \neq j$. Denote the order of $s_i s_j$ by m. Let w be the longest element in the subgroup of W generated by s_i and s_j. It has length m and exactly two reduced expressions: $w = s_i s_j \cdots$ (m factors) and $w = s_j s_i$ (m factors).

Consider first the case $m = 2$, i.e., the case $a_{ij} = 0$. Since $T_j(E_i) = E_i$ and $T_i(E_j) = E_j$, the two possible expressions $w = s_j s_i = s_i s_j$ yield the span of all $E_j^{(a)} E_i^{(b)}$ (resp. of all $E_i^{(a)} E_j^{(b)}$). Because $E_i E_j = E_j E_i$, we get in both cases the subalgebra generated by E_i and E_j.

Suppose now that $m = 3$, i.e., that $a_{ij} = -1$. The situation is symmetric in i and j, so it suffices to work with the expression $w = s_i s_j s_i$. The products in (3.2) are then all

$$E_i^{(a)} T_i(E_j^{(b)}) T_i T_j(E_i^{(c)}) = E_i^{(a)} T_i(E_j^{(b)}) E_j^{(c)}, \tag{3.3}$$

Let us write

$$E_{ij} = T_i(E_j) = E_j E_i - v^{-1} E_i E_j. \tag{3.4}$$

This explicit formula for $T_i(E_j)$ shows that all products as in (3.3) are contained in the subalgebra generated by E_i and E_j. It then suffices to show that this span is stable under right multiplication by E_i and E_j. This stability is obvious in the case of E_i; the proof for E_j uses a few commutator formulas that we are going to derive now.

First we note that by the Serre relation of E_j and E_i, we have

$$E_jE_{ij} = E_j^2E_i - v^{-1}E_jE_iE_j = E_j^2E_i - v^{-1}E_jE_iE_j = vE_jE_iE_j - E_iE_j^2 = vE_{ij}E_j$$

and hence

$$E_jE_{ij}^{(n)} = v^nE_{ij}^{(n)}E_j \quad \text{for} \quad n \geq 0.$$

On the other hand, we can rewrite (3.4) as

$$E_jE_i = v^{-1}E_iE_j + E_{ij}$$

and this yields (by induction) for all $n \geq 1$

$$E_jE_i^{(n)} = v^{-n}E_i^{(n)}E_j + [n]E_i^{(n-1)}E_{ij}.$$

Now for all $a, b, c \geq 0$, we have

$$\begin{aligned} E_j \cdot E_i^{(a)}E_{ij}^{(b)}E_j^{(c)} &= v^{-a}E_i^{(a)}E_jE_{ij}^{(b)}E_j^{(c)} + [a]E_i^{(a-1)}E_{ij}E_{ij}^{(b)}E_j^{(c)} \\ &= v^{b-a}E_i^{(a)}E_{ij}^{(b)}E_j^{(c+1)} + [a][b]E_j^{(a-1)}E_{ij}^{(b+1)}E_i^{(c)}, \end{aligned}$$

so the span of the products in (3.3) is indeed stable under left multiplication with E_i. This completes the proof. □

For any $w \in W$ denote the span of all products as in (3.3) by $\mathbf{U}^+[w]$. One can show that $\mathbf{U}^+[w]$ is a subalgebra of $\mathbf{U}^+$, but we do not need this result. Right now we need only the much weaker statement:

Lemma 3.3. *If $w \in W$ and $i \in \mathbb{I}$ with $w^{-1}(\alpha_i) < 0$, then $E_i\mathbf{U}^+[w] \subseteq \mathbf{U}^+[w]$.*

Proof. In fact, the assumption $w^{-1}(\alpha_i) < 0$ implies that we can find a reduced expression $w = s_{i_1}s_{i_2}\cdots s_{i_t}$ with $i_1 = i$. Then the products in (3.3) starts with $E_i^{(a)}$, so their span is clearly stable under left multiplication with E_i. □

The previous results applies in particular to reduced expressions of the longest element w_0 of W. In this case, the algebra $\mathbf{U}^+[w_0]$ coincides with $\mathbf{U}^+$, as shown by the following theorem:

Theorem 3.4. *Let $w_0 = s_{i_1}\cdots s_{i_\nu}$ be a reduced expression of the longest element $w_0 \in W$. Then the elements (for $\mathbf{c} \in \mathbb{N}^\nu$)*

$$E_{i_1}^{(c_1)}T_{i_1}(E_{i_2}^{(c_2)})\cdots T_{i_1}\cdots T_{i_{\nu-1}}(E_{i_\nu}^{(c_\nu)}) \tag{3.5}$$

are contained in $\mathbf{U}^+$ and form a $\mathbb{Q}(v)$-basis of $\mathbf{U}^+$.

Proof. By our convention, the products in (3.5) span $\mathbf{U}^+[w_0]$. We have $E_i\mathbf{U}^+[w_0] \subseteq \mathbf{U}^+[w_0]$ for all $i \in \mathbb{I}$ by Lemma 3.3. Since the E_i generate $\mathbf{U}^+$ and $1 \in \mathbf{U}^+[w_0]$, this implies that $\mathbf{U}^+[w_0] = \mathbf{U}^+$, i.e. the products $E_\mathbf{i}^\mathbf{c}$ generate $\mathbf{U}^+$. We have already seen that they are linearly independent, so they form a basis of $\mathbf{U}^+$. □

4 Definition of the canonical basis B

In order to properly label the basis given by Theorem 3.4, we introduce the following notations. Let $\mathscr{X}$ be the set of all sequences $\mathbf{i} = (i_1, \dots, i_\nu)$ of elements in $\mathbb{I}$ such that $s_{i_1} \cdots s_{i_\nu} = w_0$, where $\nu = \#R^+$. For each $\mathbf{i} \in \mathscr{X}$, Theorem 3.4 gives us a PBW type basis of $\mathbf{U}^+$, which will be denoted by $E_{\mathbf{i}}^{\mathbf{c}}$, $\mathbf{c} \in \mathbb{N}^\nu$. We shall regard $\mathscr{X}$ as a set of vertices of a graph $\mathbf{X}$ in which $\mathbf{i} = (i_1, \dots, i_\nu)$ and $\mathbf{i}' = (i'_1, \dots, i'_\nu)$ are joined if $\mathbf{i}'$ is obtained from $\mathbf{i}$ by an elementary move. More precisely, this happens if and only if $\mathbf{i}'$ is obtained from $\mathbf{i}$ by

(a) replacing three consecutive entries i, j, i in $\mathbf{i}$ with $a_{ij} = -1$ by j, i, j;

(b) replacing two consecutive entries i, j in $\mathbf{i}$ (with $a_{ij} = 0$) by j, i.

For such joined $(\mathbf{i}, \mathbf{i}')$ (i.e. in case (a) (resp. case (b))), we define a map $R_{\mathbf{i}}^{\mathbf{i}'} : \mathbb{N}^\nu \xrightarrow{\sim} \mathbb{N}^\nu$ as follows. The map $R_{\mathbf{i}}^{\mathbf{i}'}$ takes $\mathbf{c} = (c_1, \dots, c_\nu) \in \mathbb{N}^\nu$ to $\mathbf{c}' = (c'_1, \dots, c'_\nu) \in \mathbb{N}^\nu$, which has the same coordinates as $\mathbf{c}$ except in the three (resp. two) consecutive positions at which $(\mathbf{i}, \mathbf{i}')$ differ; if (a, b, c) (resp. (a, b)) are the coordinates of $\mathbf{c}$ at those three (resp. two) positions, the coordinates of $\mathbf{c}'$ at those positions are

$$(b + c - \min\{a, c\}, \min\{a, c\}, a + b - \min\{a, c\}) \quad (\text{resp. } (b, a)).$$

Lemma 4.1. *The map $R_{\mathbf{i}}^{\mathbf{i}'}$ is a bijection, and its inverse is $R_{\mathbf{i}'}^{\mathbf{i}}$.*

Proof. This is clear for the second case, so assume that we are in case (a). Then the map $(R_{\mathbf{i}}^{\mathbf{i}'})^2$ acts on (a, b, c) with $a < c$ via

$$(a, b, c) \mapsto (b + c - a, a, b) \mapsto (a, b, c)$$

so it is involutive. Finally note that $R_{\mathbf{i}'}^{\mathbf{i}} = R_{\mathbf{i}}^{\mathbf{i}'}$. □

As we have already remarked, any two reduced expressions of w_0 can be obtained from one to another by (a) and (b), so the graph $\mathbf{X}$ is connected. We now define a new graph $\widetilde{\mathbf{X}}$ whose vertices are pairs $(\mathbf{i}, \mathbf{c}) \in \mathscr{X} \times \mathbb{N}^\nu$, and two such pairs $(\mathbf{i}, \mathbf{c})$, $(\mathbf{i}', \mathbf{c}')$ are joined precisely when $(\mathbf{i}, \mathbf{i}')$ are joined in $\mathbf{X}$ and $R_{\mathbf{i}}^{\mathbf{i}'}(\mathbf{c}) = \mathbf{c}'$. Then there is a morphism of graphs $\mathrm{pr}_1 : \widetilde{\mathbf{X}} \to \mathbf{X}$.

Hence, given $\mathbf{i}, \mathbf{i}' \in \mathscr{X}$ and $\mathbf{c} \in \mathbb{N}^\nu$, we can write uniquely

$$E_{\mathbf{i}}^{\mathbf{c}} = \sum_{\mathbf{c}' \in \mathbb{N}^\nu} \gamma_{\mathbf{i},\mathbf{i}'}^{\mathbf{c},\mathbf{c}'} E_{\mathbf{i}'}^{\mathbf{c}'},$$

where $\gamma_{\mathbf{i},\mathbf{i}'}^{\mathbf{c},\mathbf{c}'} \in \mathbb{Q}(v)$. The advantage of introducing the operators $R_{\mathbf{i}}^{\mathbf{i}'}$ is that they encode the coefficients of $\gamma_{\mathbf{i},\mathbf{i}'}^{\mathbf{c},\mathbf{c}'}$, as shown in the following proposition:

Proposition 4.2. *Let $\mathbf{i} \in \mathscr{X}$ and $B_{\mathbf{i}}$ be the basis given in Theorem 3.4.*

(a) $B_{\mathbf{i}}$ is a basis of U^+ as an $\mathbb{Z}[v, v^{-1}]$-module.

(b) Let $\mathscr{L}_{\mathbf{i}}$ be the $\mathbb{Z}[v^{-1}]$-submodule of U^+ generated by the basis $B_{\mathbf{i}}$. Then $\mathscr{L}_{\mathbf{i}}$ is independent of $\mathbf{i}$; we denote it by $\mathscr{L}$.

(c) Let $\pi : \mathscr{L} \to \mathscr{L}/v^{-1}\mathscr{L}$ the canonical projection. Then $\pi(B_{\mathbf{i}})$ is a basis of $\mathscr{L}/v^{-1}\mathscr{L}$, independent of $\mathbf{i}$; we denote it by B.

(d) Assume that $\mathbf{i}, \mathbf{i}' \in \mathscr{X}$ are joined in the graph $\mathbf{X}$. For $\mathbf{c}, \mathbf{c}' \in \mathbb{N}^\nu$, $\gamma_{\mathbf{i},\mathbf{i}'}^{\mathbf{c},\mathbf{c}'}$ is in $\mathbb{Z}[v^{-1}]$. Its constant term is 1 if $\mathbf{c}' = R_{\mathbf{i}}^{\mathbf{i}'}(\mathbf{c})$ and zero otherwise.

Proof. Assume that (d) holds. To prove that the objects defined in (b),(c) in terms of $\mathbf{i}, \mathbf{i}' \in \mathscr{X}$ coincide, we may assume that $\mathbf{i}, \mathbf{i}'$ are joined in $\mathscr{X}$, in which case the desired statements follow immediately from (d). Let's first show that $A\mathscr{L} = U^+$. It is enough to prove that, for any i and N, $A\mathscr{L}$ is stable under left multiplication by $E_i^{(N)}$. But if we choose (as we may) $\mathbf{i} \in \mathscr{X}$ with first entry i, then $A\mathscr{L}_{\mathbf{i}}$ is clearly stable under left multiplication by $E_i^{(N)}$. Since $A\mathscr{L}_{\mathbf{i}} = A\mathscr{L}$ by (b), we see that (a) holds.

It remains to prove (d). Using the definitions and braid group relation, we see that it is enough to verify (d) under the assumption that $n = 2$. In the remainder of this proof we shall therefore assume that $n = 2$. The case where $a_{ij} = 0$ is trivial, we shall therefore assume that $a_{ij} = -1$. In this case we set $\mathbf{i} = (i, j, i)$, $\mathbf{i}' = (j, i, j)$. By definition, we have

$$T_i^{-1}(E_j) = E_jE_i - v^{-1}E_iE_j = T_j(E_i), \quad T_i^{-1}T_j^{-1}(E_i) = T_i^{-1}T_i(E_j) = E_j,$$

so for $\mathbf{c} = (c_1, c_2, c_3) \in \mathbb{N}^3$ and $\mathbf{c}' = (c_1', c_2', c_3') \in \mathbb{N}^3$, we have

$$\begin{aligned} E_{\mathbf{i}}^{\mathbf{c}} &= E_i^{(c_1)}(E_jE_i - v^{-1}E_iE_j)^{(c_2)}E_j^{(c_3)}, \\ E_{\mathbf{i}'}^{\mathbf{c}'} &= E_j^{(c_1')}(E_iE_j - v^{-1}E_jE_i)^{(c_2')}E_i^{(c_3')}. \end{aligned}$$

Let $(p, q, r) \in \mathbb{N}^3$ be such that $q \geq p + r$. We shall make use the following identity for the monomial $E_i^{(p)}E_j^{(q)}E_i^{(r)}$, which can be found in [Lus10, Lemma 42.1.2]:

$$E_i^{(p)}E_j^{(q)}E_i^{(r)} = \sum_{s=0}^{r} v^{-(q-s)(r-s)} \begin{bmatrix} p-s+r \\ r-s \end{bmatrix} E_{\mathbf{i}}^{(p-s+r,s,q-s)},$$

where

$$v^{-(q-s)(r-s)} \begin{bmatrix} r-s+p \\ r-s \end{bmatrix} \in v^{-(r-s)(q-p-r)}(1 + v^{-1}\mathbb{Z}[v^{-1}])$$

is in $v^{-1}\mathbb{Z}[v^{-1}]$ if $s < r$ and equals to 1 if $s = r$. Similarly, we have

$$E_j^{(p)}E_i^{(q)}E_j^{(r)} = \sum_{s=0}^{p} v^{-(p-s)(q-s)} \begin{bmatrix} p-s+r \\ p-s \end{bmatrix} E_{\mathbf{i}}^{(q-s,s,p-s+r)},$$

where

$$v^{-(p-s)(q-s)} \begin{bmatrix} p-s+r \\ p-s \end{bmatrix} \in v^{-(p-s)(q-r-s)}(1 + v^{-1}\mathbb{Z}[v^{-1}])$$

is in $v^{-1}\mathbb{Z}[v^{-1}]$ if $s < p$ and equals to 1 if $s = p$. Now consider the family $\mathscr{F}$ consisting of the following elements of U^+:

$$E_i^{(p)}E_j^{(q)}E_i^{(r)}, \tag{4.1}$$
$$E_j^{(p)}E_i^{(q)}E_j^{(r)} \tag{4.2}$$

for various $(p, q, r) \in \mathbb{N}^3$ such that $q \geq p + r$, where elements

$$E_i^{(p)}E_j^{(p+r)}E_i^{(r)} = E_j^{(r)}E_i^{(p+r)}E_j^{(p)}$$

which are both of type (4.1) and (4.2), are considered only once. Let $\mathscr{L}$ be the $\mathbb{Z}[v^{-1}]$ submodule of U^+ generated by $\mathscr{F}$. We have a bijection $\sigma : \mathscr{F} \xrightarrow{\sim} B_{\mathbf{i}}$ so that to an element (4.1) we associate $E_{\mathbf{i}}^{(p,r,q-r)}$ and to an element (4.2) we associate $E_{\mathbf{i}}^{(q-p,p,r)}$. The previous computations show that each element in $\mathscr{F}$ is equal to the corresponding element of form $E_{\mathbf{i}}^{(a,b,c)}$ plus a linear combination with coefficients in $v^{-1}\mathbb{Z}[v^{-1}]$ of elements of the form $E_{\mathbf{i}}^{(a',b',c')}$ with $b' < b$,

$a+b = a'+b'$, $b+c = b'+c'$. Hence $\mathscr{L} = \mathscr{L}_{\mathbf{i}}$, $\mathscr{F}$ is a $\mathbb{Z}[v^{-1}]$-basis of $\mathscr{M}$ and the image of $\mathscr{F}$ under the canonical image $\pi : \mathscr{L} \to \mathscr{L}/v^{-1}\mathscr{L}$ coincides with the image of $B_{\mathbf{i}}$. The automorphism exchanging E_i and E_j of $\mathbf{U}^+$ leaves $\mathscr{F}$, $\mathscr{L}$ stable, and takes $\mathscr{L}_{\mathbf{i}}$ onto $\mathscr{L}_{\mathbf{i}'}$ and $E_{\mathbf{i}}^{(a,b,c)}$ to $E_{\mathbf{i}'}^{(a,b,c)}$. Hence we have $\mathscr{L}_{\mathbf{i}} = \mathscr{L}_{\mathbf{i}'} = \mathscr{L}$ and the images of $B_{\mathbf{i}}$ and $B_{\mathbf{i}'}$ under $\pi : \mathscr{L} \to \mathscr{L}/v^{-1}\mathscr{L}$ coincide. We obtain a bijection between $B_{\mathbf{i}}$ and $B_{\mathbf{i}'}$ as follows: $E_{\mathbf{i}}^{(a,b,c)}$ corresponds to $E_{\mathbf{i}'}^{(a',b',c')}$ precisely when they have the same image under $\pi : \mathscr{L} \to \mathscr{L}/v^{-1}\mathscr{L}$. This image is then the same as that of an element β as in (4.1) or (4.2). Assume that β is as in (4.1), then as we have seen, we have $(a,b,c) = (p,r,q-r)$ and similarly $(a',b',c') = (q-p,p,r)$. Hence

$$(a',b',c') = (b+c-\min\{a,c\}, \min\{a,c\}, a+b-\min\{a,c\}) \tag{4.3}$$

Assume next that β is as in (4.2). Then as we have seen we have $(a,b,c) = (q-p,p,r)$ and $(a',b',c') = (p,r,q-r)$. Hence (4.3) still holds. This completes the proof of the proposition. □

Let $\widehat{\mathscr{X}}$ be the set of all morphisms of graphs $\sigma : \mathbf{X} \to \widetilde{\mathbf{X}}$ such that $\mathrm{pr}_1 \circ \sigma = 1$. We may regard $\widehat{\mathscr{X}}$ as the set of all maps $\phi : \mathscr{X} \to \mathbb{N}^\nu$ such that $R_{\mathbf{i}}^{\mathbf{i}'}(\phi(\mathbf{i})) = \phi(\mathbf{i}')$ whenever $(\mathbf{i},\mathbf{i}')$ are joined in $\mathbf{X}$. We can define a map

$$\Phi : B \to \widehat{\mathscr{X}} \tag{4.4}$$

as follows. To $b \in B$, we associate the function $\phi : \mathscr{X} \to \mathbb{N}^\nu$ given by $\phi(\mathbf{i}) = \mathbf{c}$ where $\pi(E_{\mathbf{i}}^{\mathbf{c}}) = b$. From Proposition 4.2, it follows immediately that $\phi \in \widehat{\mathscr{X}}$.

For any $\mathbf{i} \in \mathscr{X}$, we have a map $\mathrm{ev}_{\mathbf{i}} : \widehat{\mathscr{X}} \xrightarrow{\sim} \mathbb{N}^\nu$ given by $\phi \mapsto \phi(\mathbf{i})$ and a bijection $\mathbb{N}^\nu \cong B$ defined by $\mathbf{c} \mapsto \pi(E_{\mathbf{i}}^{\mathbf{c}}) \in B$. Their composition is a map $\widehat{\mathscr{X}} \to B$, and it is clear that this is the inverse of the map Φ. Thus we have

Corollary 4.3. *The map $\Phi : B \to \widehat{\mathscr{X}}$ is a bijeciton.*

Let $\mathbf{i}, \mathbf{i}'$ be two elements of $\mathscr{X}$. We define a bijection $R_{\mathbf{i}}^{\mathbf{i}'} : \mathbb{N}^\nu \xrightarrow{\sim} \mathbb{N}^\nu$ as the composition

$$R_{\mathbf{i}}^{\mathbf{i}'} = R_{\mathbf{i}_{p-1}}^{\mathbf{i}_p} \circ R_{\mathbf{i}_{p-2}}^{\mathbf{i}_{p-1}} \cdots \circ R_{\mathbf{i}_1}^{\mathbf{i}_2}$$

where $(\mathbf{i} = \mathbf{i}_1, \mathbf{i}_2, \ldots, \mathbf{i}_p = \mathbf{i}')$ is a path in $\mathbf{X}$ (we use the connectedness of $\mathbf{X}$). This bijection is independent of the choice of path as it is equal to the composition of the bijection $\mathrm{ev}_{\mathbf{i}}$ with the inverse of the bijection $\mathrm{ev}_{\mathbf{i}'}$.

Let $\mathbf{i} \in \mathscr{X}$. We associate to $\mathbf{i}$ the sequence $\beta_1, \ldots, \beta_\nu$ defined by

$$\beta_k = s_{i_1} s_{i_2} \cdots s_{i_{k-1}}(\alpha_{i_k}) \quad \text{for} \quad k = 1, \ldots, \nu.$$

This sequence contains each root in R^+ exactly once, so we can write uniquely

$$\beta_k = \sum_{j=1}^{n} p_j^k \alpha_j$$

where $p_j^k \in \mathbb{N}$. We now define a map $\chi_{\mathbf{i}} : \mathbb{N}^\nu \to \mathbb{N}^{\mathbb{I}}$ by

$$\mathbf{c} = (c_1, \ldots, c_\nu) \mapsto \mu = \sum_{i=1}^{n} \mu_i \alpha_i \tag{4.5}$$

where

$$\mu_j = \sum_{k=1}^{\nu} p_j^k c_k.$$

The fibres of this map are finite. An element of $\mathbb{N}^\nu$ in the fibre of $\chi_{\mathbf{i}}$ over μ is said to have **i-homogeneity** μ. If $\mathbf{i} \in \mathscr{X}$, the basis $B_{\mathbf{i}}$ consists of homogeneous elements. The part of this basis that is contained in $\mathbf{U}_\mu^+$ corresponds under the bijection $\pi : B_{\mathbf{i}} \to B$ to the inverse image of μ under the map $\chi_{\mathbf{i}}$.

Remark 4.4. Let $\mathbf{i} = (i_1, \dots, i_\nu) \in \mathscr{X}$ and $\mathbf{i}' = (i'_1, \dots, i'_\nu)$ be defined by $s_{i'_j} = w_0 s_{i_{\nu-j+1}} w_0^{-1}$. Then $\mathbf{i}' \in \mathscr{X}$, and one verified easily that for any $k \in [1, \nu]$ we have

$$s_{i_{k-1}} s_{i_{k-2}} \cdots s_{i_1} s_{i'_1} s_{i'_2} \cdots s_{i'_{\nu-k}} s_{i'_{\nu-k+1}} = w_0$$

and[2]

$$s_{i_{k-1}} s_{i_{k-2}} \cdots s_{i_1} s_{i'_1} s_{i'_2} \cdots s_{i'_{\nu-k}} (\alpha_{i'_{\nu-k+1}}) = \alpha_{i_k}.$$

Hence, using Proposition 2.2, we have

$$T_{i_{k-1}} T_{i_{k-2}} \cdots T_{i_1} T_{i'_1} T_{i'_2} \cdots T_{i'_{\nu-k}} (E_{i'_{\nu-k+1}}) = E_{i_k}.$$

It follows that

$$T_{i'_1} T_{i'_2} \cdots T_{i'_{\nu-k}} (E_{i'_{\nu-k+1}}) = T_{i_1} T_{i_2} \cdots T_{i_{k-1}} (E_{i_k}) = \tau(T_{i_1} T_{i_2} \cdots T_{i_{k-1}} (E_{i_k})).$$

This identity shows that $\tau(E_{\mathbf{i}}^{\mathbf{c}}) = E_{\mathbf{i}'}^{\mathbf{c}'}$, where $\mathbf{c}' = (c'_1, \dots, c'_\nu)$ is related to $\mathbf{c} = (c_1, \dots, c_\nu)$ by $c'_j = c_{\nu-j+1}$. In particular, we have

$$\tau(B_{\mathbf{i}}) = B_{\mathbf{i}'}, \quad \tau(\mathscr{L}) = \mathscr{L}. \tag{4.6}$$

Applying the bar involution to $\mathscr{L}$, we obtain a $\mathbb{Z}[v]$-submodule $\overline{\mathscr{L}}$ of U^+. The canonical basis of $\mathbf{U}$ is then defined to be certain bar-invariant elements in the intersection $\mathscr{L} \cap \overline{\mathscr{L}}$. Its elementary properties are described by the following theorem.

Theorem 4.5. (a) *The restriction of $\pi : \mathscr{L} \to \mathscr{L}/v^{-1}\mathscr{L}$ defines an isomorphism of $\mathbb{Z}$-modules $\tilde{\pi} : \mathscr{L} \cap \overline{\mathscr{L}} \xrightarrow{\sim} \mathscr{L}/v^{-1}\mathscr{L}$. Hence, if we set $\mathbf{B} = \tilde{\pi}^{-1}(B)$, then $\mathbf{B}$ is a $\mathbb{Z}$-basis of $\mathscr{L} \cap \overline{\mathscr{L}}$.*

(b) *$\mathbf{B}$ is a $\mathbb{Z}[v^{-1}]$-basis of $\mathscr{L}$, a $\mathbb{Z}[v]$-basis of $\overline{\mathscr{L}}$, an A-basis of U^+ and a $\mathbb{Q}(v)$-basis of $\mathbf{U}^+$.*

(c) *Each element of $\mathbf{B}$ is bar invariant.*

The proof will be given in §8. We call $\mathbf{B}$ the *canonical basis*.

Proposition 4.6. *We have $\tau(\mathbf{B}) = \mathbf{B}$.*

Proof. By (4.6), τ maps $\mathscr{L}$ onto itslf; it induces an automorphism of $\mathscr{L}/v^{-1}\mathscr{L}$ that maps B onto itself, as we see from (4.6). Clearly, τ commutes with bar involution and with $\pi : \mathscr{L} \to \mathscr{L}/v^{-1}\mathscr{L}$, hence leaves stable $\mathscr{L} \cap \overline{\mathscr{L}}$ and also $\mathbf{B}$. The proposition follows. □

It is clear that by exchanging the role of E_i and F_i, we can define a canonical basis for the negative part $\mathbf{U}^-$, and they satisfy similar properties to $\mathbf{B}$. It is worth noting that the canonial basis $\mathbf{B}$ is "almost orthogonal" with respect to a pairing of $\mathbf{U}^+$. For details, one may consult [Lus10].

5 Quivers and BGP reflection functors

In this section we shall recall some basic facts on quivers and the BGP functors. Recall that the Dynkin graph of (a_{ij}) has $\mathbb{I} = [1, n]$ as set of vertices and i, j form an edge precisely when $a_{ij} = -1$. Assume given an orientation Ω of this Dynkin graph, i.e., a collection of arrows $i \to j$, one for each edge i, j in the Dynkin graph. For an edge h in Ω, we denote by $s(h)$ its source and $t(h)$ its target.

[2] Note that by definition we have $s_{i_{\nu-j+1}} w_0^{-1}(\alpha_{i'_j}) = -w_0^{-1}(\alpha_{i'_j})$, so $w_0^{-1}(\alpha_{i'_j}) = -\alpha_{i_{\nu-j+1}}$.

Let $\Bbbk$ be a fixed field and $\mathbf{Mod}(Q)$ be the category of finitely generated modules of the quiver $Q = (\mathbb{I}, \Omega)$. An object of $\mathbf{Mod}(Q)$ is a finite-dimensional $\mathbb{I}$-graded $\Bbbk$-vector space $V = \bigoplus_i V_i$ and $\Bbbk$-linear maps $x_h : V_{s(h)} \to V_{t(h)}$ for an edge $h \in \Omega$. A morphism of this object to another object (W, y_h) is a collection of $\Bbbk$-linear maps $g_i : V_i \to W_i$ such that $y_h g_{s(h)} = g_{t(h)} x_h$ for all $h \in \Omega$. Then $\mathbf{Mod}(Q)$ is an abelian category in an obvious way.

The dimension of $V = \bigoplus_i V_i$ is the sequence

$$\mathbf{dim}(V) = (\dim(V_i)) \in \mathbb{N}^{\mathbb{I}}.$$

For $i \in \mathbb{I}$, let $S(i)$ be the module defined by $V_i = \Bbbk$ and $V_j = 0$ for $j \neq i$. This module is simple and any simple module is isomorphic to an $S(i)$ for a unique i.

Assume that $i \in \mathbb{I}$ is a sink (resp. a source) of Q, i.e. that there is no arrow $i \to j$ (resp. $j \to i$) in Ω. Let $\mathbf{Mod}_i^+(Q)$ (resp. $\mathbf{Mod}_i^-(Q)$) denote the full subcategory of $\mathbf{Mod}(Q)$ whose objects are (V, x) such that $\bigoplus_{t(h)=i} x_h : \bigoplus_{t(h)=i} V_{s(h)} \to V_i$ is surjective (resp. $\bigoplus_{s(h)=i} x_h : V_i \to \bigoplus_{s(h)=i} V_{t(h)}$ is injective). The following statement is easily verified:

Proposition 5.1. *Let V be a module in $\mathbf{Mod}(Q)$. We have $V \in \mathbf{Mod}_i^+(Q)$ (resp. $V \in \mathbf{Mod}_i^-(Q)$) if and only if* $\mathrm{Hom}_{\Bbbk Q}(V, S(i)) = 0$ *(resp.* $\mathrm{Hom}_{\Bbbk Q}(S(i), V) = 0$*).*

Let $s_i\Omega$ be the orientation obtained from Ω by reversing each arrow that ends (resp. starts) at i; then i is a source (resp. a sink) of $s_i\Omega$. We denote by s_iQ the quiver with this new orientation. Following [BGP73], we define the reflection functor $\Phi_i^+ : \mathbf{Mod}(Q) \to \mathbf{Mod}_i^-(s_iQ)$ (resp. $\Phi_i^- : \mathbf{Mod}(Q) \to \mathbf{Mod}_i^+(s_iQ)$) by associating to an object V in $\mathbf{Mod}(Q)$ the object V' in $\mathbf{Mod}(s_iQ)$ defined by

$$V_i' = \ker\Big(\bigoplus_{t(h)=i} x_h : \bigoplus_{t(h)=i} V_{s(h)} \to V_i\Big)$$

and $V_k' = V_k$ for $k \neq i$ (resp.

$$V_i' = \mathrm{coker}\Big(\bigoplus_{s(h)=i} x_h : V_i \to \bigoplus_{s(h)=i} V_{t(h)}\Big)$$

and $V_k' = V_k$ for $k \neq i$); the maps x_h' are the obvious ones. On morphisms, Φ_i^+ (resp. Φ_i^-) is defined in the obvious way.

Assume that i is a sink of Q. For any module V in $\mathbf{Mod}(Q)$ there is a canonical exact sequence

$$0 \longrightarrow \Phi_i^-\Phi_i^+(V) \to V \to V(i) \to 0 \tag{5.1}$$

where $V(i)$ has i-component $\mathrm{coker}(\bigoplus_{t(h)=i} x_h : \bigoplus_{t(h)=i} V_{s(h)} \to V_i)$ and its other components are 0. In particular,

$$\Phi_i^-\Phi_i^+(V) \cong V \iff V \in \mathbf{Mod}_i^+(Q) \tag{5.2}$$

and the restriction of Φ_i^+ defines an equivalence of categories

$$\mathbf{Mod}_i^+(Q) \xrightarrow{\sim} \mathbf{Mod}_i^-(s_iQ) \tag{5.3}$$

whose inverse is given by the restriction of Φ_i^-. If V in $\mathbf{Mod}_i^+(Q)$ corresponds to V' in $\mathbf{Mod}_i^-(s_iQ)$ under this equivalence, then we have

$$\mathbf{dim}(V') = s_i(\mathbf{dim}(V)). \tag{5.4}$$

We also note that following theorem for the functors Φ_i^+ and Φ_i^-. Here we use notation $R^n\Phi$ for the derived functors of a left exact functor Φ, and $L^n\Phi$ for the derived functors of a right exact functor Φ. Its proof can be found for example, in [Kir16].

Theorem 5.2. *Let i be a sink (resp. a source). Then the functor Φ_i^+ (resp. Φ_i^-) is left exact (resp. right exact). Moreover, $R^n\Phi_i^+(M) = 0$ (resp. $L^n\Phi_i^-(M) = 0$) for all $n > 1$, and $R^1\Phi_i^+(M) = 0$ (resp. $L^1\Phi_i^-(M) = 0$) iff M is in $\mathbf{Mod}_i^+(Q)$ (resp. $\mathbf{Mod}_i^-(Q)$).*

A key application of the reflection functors Φ_i^+ and Φ_i^- is that they can be used to construct all representations of the quiver Q, each corresponding to a dimension vector in the root system R. For this, it is convenient to introduce the following definition:

Definition 5.3. Let M, M', M'' be modules in $\mathbf{Mod}(Q)$; assume that the following conditions are satisfied:

(a) M is isomorphic to $M' \oplus M''$,

(b) any exact sequence $0 \to M' \to N \to M'' \to 0$ in $\mathbf{Mod}(Q)$ splits,

(c) there is a unique submodule of M that is isomorphic to M' and is such that the quotient of M by this submodule is isomorphic to M''.

We express this by the notation $M \cong M' * M''$. For example, in (5.1), we have

$$V \cong V(i) * \Phi_i^-\Phi_i^+(V). \tag{5.5}$$

Proposition 5.4. *Assume that i is a sink of Q and that $M', M'' \in \mathbf{Mod}_i^+(Q)$; let $\overline{M}'$, $\overline{M}''$ be the corresponding objects in $\mathbf{Mod}_i^-(s_iQ)$ (under (5.3)). Assume also that $\overline{M} \in \mathbf{Mod}(s_iQ)$ satisfies $\overline{M} \cong \overline{M}' * \overline{M}''$. Then $\overline{M} \in \mathbf{Mod}_i^-(s_iQ)$ and the corresponding object (under (5.3)) $M \in \mathbf{Mod}_i^+(Q)$ satisfies $M \cong M' * M''$.*

Proof. From our definition, it is clear that $\overline{M}$ is in $\mathbf{Mod}_i^-(s_iQ)$. We also note that the subcategories $\mathbf{Mod}_i^+(Q)$ and $\mathbf{Mod}_i^-(s_iQ)$ are stable under extensions. Since Φ_i^+ restricts to an equivalence, it is easy to see that conditions (b) and (c) holds for M, so the proposition follows. □

Example 5.5. Assume that i is a sink of Q and that V is an indecomposable module in $\mathbf{Mod}(Q)$, not isomorphic to any $S(i)$. From (5.5) it follows that V is in $\mathbf{Mod}_i^+(Q)$. It also follows that (5.3) defines a bijection between the following two sets:

(a) the set of indecomposable modules (up to isomorphism) other than $S(i)$ in $\mathbf{Mod}_i^+(Q)$ and

(b) the set of indecomposable modules (up to isomorphism) other than $S(i)$ in $\mathbf{Mod}_i^-(s_iQ)$.

On the other hand, it is clear that $\Phi_i^+(S(i)) = 0$.

Now assume that we are given an element $\mathbf{i} = (i_1, \ldots, i_\nu) \in \mathcal{X}$. We say that $\mathbf{i}$ is *adapted* to the orientation Ω if for each k, i_k is a sink of $Q_k := s_{i_{k-1}} \cdots s_{i_2}s_{i_1}Q$. If V is a module in $\mathbf{Mod}(Q)$, $k \in [0, \nu]$ and $l \in [1, k+1]$, we set ($V^{k+1,k} = 0$)

$$V^k = \Phi_{i_k}^+ \cdots \Phi_{i_2}^+\Phi_{i_1}^+(V) \in \mathbf{Mod}(Q_{k+1}), \quad V^{l,k} = \Phi_{i_l}^- \cdots \Phi_{i_{k-1}}^-\Phi_{i_k}^-(V^k) \in \mathbf{Mod}(Q_l). \tag{5.6}$$

Lemma 5.6. *We have $V^\nu = 0$.*

Proof. In fact, assume that V^ν is nonzero for some V. We may assume that V is indecomposable. From Example 5.5, it follows that $V \in \mathbf{Mod}_{i_1}^+(Q_1)$, $\Phi_{i_1}^+V \in \mathbf{Mod}_{i_2}^+(Q_2)$, $\Phi_{i_2}^+\Phi_{i_1}^+V \in \mathbf{Mod}_{i_3}^+(Q_3)$, etc. Now using (5.4) we see that

$$\mathbf{dim}(V^\nu) = s_{i_\nu} \cdots s_{i_2}s_{i_1}\mathbf{dim}(V) = w_0^{-1}\mathbf{dim}(V) = w_0\mathbf{dim}(V).$$

In the left-hand side of this equality the coefficients are in $\mathbb{N}$ and not all are zero; in the right-hand side the coefficients are in $-\mathbb{N}$; this is a contradiction. □

Lemma 5.7. *Let M_k be a module in $\mathbf{Mod}(Q_k)$ whose component at any vertex $\neq i_k$ is zero; assume that $2 \leq l \leq k$. Then the module $P' = \Phi^-_{i_l} \cdots \Phi^-_{i_{k-2}} \Phi^-_{i_{k-1}}(M_k)$ is in $\mathbf{Mod}^-_{i_{l-1}}(Q_l)$.*

Proof. Indeed, we can assume that $M_k = S(i_k)$. From Example 5.5 we see that P' is indecomposable. Let $\mathbf{d} = \mathbf{dim}(P') \in \mathbb{Z}^{\mathbb{I}}$. If P' is not in $\mathbf{Mod}^-_{i_{l-1}}(Q_l)$ then $P' \cong S(i_{l-1})$ (again by Example 5.5) and then $s_{i_{l-1}}(\mathbf{dim}(P')) \in \mathbb{Z}^n$ would have some strictly negative coordinate. On the other hand, this vector is equal to $s_{i_{l-1}} s_{i_l} \cdots s_{i_{k-2}} s_{i_{k-1}}(\alpha_{i_k})$, which has positive coordinates since $s_{i_{l-1}} \cdots s_{i_{k-2}} s_{i_{k-1}} s_{i_k}$ is a reduced expression in W ([Bou02, VI, §1, no.4, Prop. 17]). This contradiction proves the assertion. ☐

Proposition 5.8. *Assume that $\mathbf{i} \in \mathscr{X}$ is adapted to Ω and V is in $\mathbf{Mod}(Q)$.*

(a) *We have a canonical filtration in $\mathbf{Mod}(Q)$:*

$$V = V^{1,0} \supset V^{1,1} \supset \cdots \supset V^{1,\nu-1} \supset V^{1,\nu} = 0.$$

Moreover $V^{1,k-1}/V^{1,k} = P_k$ where

$$P_k = \Phi^-_{i_1} \Phi^-_{i_2} \cdots \Phi^-_{i_{k-1}}(M_k)$$

and $M_k = V^{k-1}/V^{k,k}$ is a module in $\mathbf{Mod}(Q_k)$ whose j-component is zero for any $j \neq i_k$ (notation of (5.6)).

(b) *We have $V^{1,k-1} \cong P_k * V^{1,k}$ for any $k \in [1,\nu]$, hence*

$$V = P_1 * (P_2 * (\cdots * P_\nu)).$$

(c) *If $l < k$, we have $\mathrm{Hom}_{\Bbbk Q}(P_k, P_l) = 0$.*

Proof. From (5.1) we have for $k \in [1,\nu]$ a canonical exact sequence in $\mathbf{Mod}(Q_k)$:

$$0 \longrightarrow V^{k,k} \longrightarrow V^{k-1} \longrightarrow M_k \longrightarrow 0 \tag{5.7}$$

where M_k is concentrated at the vertex i_k. Applying Lemma 5.7 and (5.2) repeatedly, we deduce that for any $l \in [1,k]$ we have a canonical exact sequence

$$0 \longrightarrow V^{l,k} \longrightarrow V^{l,k-1} \longrightarrow \Phi^-_{i_l} \Phi^-_{i_{l+1}} \ldots \Phi^-_{i_{k-1}}(M_k) \longrightarrow 0 \tag{5.8}$$

We will show that

$$V^{l,k-1} \cong \Phi_{i_l} \Phi^-_{i_{l+1}} \cdots \Phi^-_{i_{k-1}}(M_k) * V^{l,k} \tag{5.9}$$

for any $k \in [1,\nu]$, $l \in [1,k]$. For $l = k$ this follows from (5.5). Assume that (5.9) is known for some $2 \leq l \leq k$, we shall deduce from this the analogous statement for $l-1, k$ by applying to it $\Phi^-_{i_{l-1}}$. In view of Proposition 5.4, it suffices to verify that

$$V^{l,k} \in \mathbf{Mod}^-_{i_{l-1}}(Q_l), \quad \Phi^-_{i_l} \Phi^-_{i_{l+1}} \cdots \Phi^-_{i_{k-1}}(M_k) \in \mathbf{Mod}^-_{i_{l-1}}(Q_l).$$

The last inclusion follows from Lemma 5.7. Applying (5.8) and Theorem 5.2 repeatedly, we see that $V^{l,k}$ is a submodule of $V^{l,l-1} = V^{l-1}$, which, by definition, is in $\mathbf{Mod}^-_{i_{l-1}}(Q_l)$; hence so is any submodule of it. Thus, (5.9) is proved. We now take $l = 1$ in (5.9) and we see that (a), (b) hold.

We now prove (c). Assume that $\mathrm{Hom}_{M(Q)}(P_k, P_l) \neq 0$ for some $l < k$. Applying successively the functors $\Phi^+_{i_1}, \Phi^+_{i_2}, \ldots, \Phi^+_{i_{l-1}}$ and using (5.3), we deduce that there exists a nonzero element $\zeta \in \mathrm{Hom}(P, X_l)$, where

$$P = \Phi^-_{i_l} \Phi^-_{i_{l+1}} \cdots \Phi^-_{i_{k-1}}(M_k) \in \mathbf{Mod}(Q_l).$$

This contradicts Proposition 5.1 since $P \in \mathbf{Mod}^+_{i_l}(Q_l)$. This contradiction proves (c). ☐

Proposition 5.9. *We fix an orientation Ω of Q.*

(a) *For any $\alpha \in R^+$, there is a unique indecomposable module (up to isomorphism) $S(\alpha) \in \mathbf{Mod}(Q)$ such that $\dim(S(\alpha)) = \alpha$; any indecomposable module is isomorphic to $S(\alpha)$ for a unique α. In particular, we have $S(\alpha_i) = S(i)$.*

(b) *There exists some $\mathbf{i} \in \mathscr{X}$ adapted to Ω.*

(c) *There exists a total order $\beta_1, \dots, \beta_\nu$ in R^+ such that $\mathrm{Hom}_{\Bbbk Q}(S(\beta_k), S(\beta_l)) = 0$ for any $k > l$.*

Proof. We first show that (b) implies (a). Let $\mathbf{i}$ be adapted to Ω. From Proposition 5.8 we see that for $k \in [1, \nu]$, $\Phi^-_{i_1}\Phi^-_{i_2} \cdots \Phi^-_{i_{k-1}}(S(i_k))$ is an indecomposable module in $\mathbf{Mod}(Q)$ of dimension

$$s_{i_1}s_{i_2} \cdots s_{i_{k-1}}(\alpha_{i_k}). \tag{5.10}$$

From Proposition 5.8 (b) we see that we see that any indecomposable module in $\mathbf{Mod}(Q)$ is of the form above. Since any $\alpha \in R^+$ is of the form (5.10) for a unique $k \in [1, \nu]$, we see that (a) holds. Note that (b) implies (c) by Proposition 5.9 (c).

We now show that (a) and (c) imply (b). Consider a total order as in (c). It is clear from the definition that $S(\beta_1)$ is necessarily a simple module (hence $\beta_1 = \alpha_{i_1}$ for some $i \in \mathbb{I}$) and that i_1 is necessarily a sink of Q. The sequence $s_{i_1}(\beta_2), s_{i_1}(\beta_3), \dots, s_{i_1}(\beta_\nu), \beta_1$ plays the same role for $s_{i_1}Q$ as a $\beta_1, \dots, \beta_\nu$ for Q; the indecomposable modules in $\mathbf{Mod}(s_{i_1}Q)$ corresponding to the terms of this sequence are $\Phi^+_{i_1}(S(\beta_2)), \dots, \Phi^+_{i_1}(S(\beta_\nu)), S(\alpha_{i_1})$ (we use Example 5.5 and Proposition 5.1). By the previous argument, we see that $s_{i_1}(\beta_2) = \alpha_{i_2}$ where $i_2 \in \mathbb{I}$ is a sink of $s_{i_1}(Q)$. Continuing in this way, we find a sequence $(i_1, i_2, \dots, i_\nu)$ in $\mathbb{I}$ such that i_k is a sink of $s_{i_1}s_{i_2} \cdots s_{i_{k-1}}Q$ for $1 \le k \le \nu$ and $s_{i_1}s_{i_2} \cdots s_{i_{k-1}}(\alpha_{i_k}) = \beta_k$ for $1 \le k \le \nu$. The last condition implies that $s_{i_1}s_{i_2} \cdots s_{i_\nu}$ is a reduced expression for w_0 ([Bou02, VI, §1, no.4, Cor.2 of Prop. 17]), hence $(i_1, \dots, i_\nu)$ is adapted to Ω and (b) holds.

Next we assume that our (oriented) Dynkin graph is embedded as a subgraph of a larger (oriented) Dynkin graph and that the proposition is already known for the larger graph. Clearly, the indecomposable modules for the smaller graph may be identified with the indecomposable modules for the larger graph, which are zero at vertices outside the smaller graph. Hence (a) for the larger graph implies (a) for the smaller graph. Similarly, (c) for the larger graph implies (c) for the smaller graph: we consider an order as in (c) for the larger graph and we discard the roots that do not belong to the smaller graph. We find an order as in (c) for the smaller graph. By an earlier part of the argument, (b) for the smaller graph follows from (a) and (c) for the same graph. We can choose the imbedding above so that the larger graph has the following property: the longest element in the Weyl group is central. Thus, we have reduced the general case to the case where w_0 is central in W. In this case, we can define a Coxeter element $c \in W$ (as in [BGP73]) by $c = s_{i_1}s_{i_2} \cdots s_{i_n}$ where $(i_1, \dots, i_n)$ is a permutation of $\{1, \dots, n\}$ such that $i_j \to i_k$ in Q implies $k < j$. Then c has even order h and $c^{h/2} = w_0$. For $1 \le k \le \nu = nh/2$, we set $i_k = i_j$, where $k \equiv j \bmod n$, $1 \le j \le n$. It is easy to check that $(i_1, \dots, i_\nu)$ is adapted to $\mathbf{Mod}(Q)$, hence (b) holds. As we have seen earlier, this implies that (a) and (c) hold also. This completes the proof. $\square$

The previous proof shows that there is a one-to-one correspondence between the sequences $\mathbf{i} \in \mathscr{X}$ that are adapted to Ω and the total orders on R^+ as in Proposition 5.9 (c). This is obtained by attaching to $\mathbf{i}$ the ordering

$$\beta_1, \beta_2, \dots, \beta_\nu \tag{5.11}$$

of R^+ given by $\beta_k = s_{i_1}s_{i_2} \cdots s_{i_{k-1}}(\alpha_{i_k})$. One can show that a sequence $\mathbf{i} \in \mathscr{X}$ can be adapted to at most one orientation (this follows from Corollary 5.10).

Corollary 5.10. *Assume that $\mathbf{i} = (i_1, \dots, i_\nu) \in \mathscr{X}$ is adapted to Ω. Let $\mathbf{i}' = (i_2, \dots, i_\nu, j)$ be defined by $s_j = w_0 s_{i_1} w_0^{-1}$.*

(a) $\mathbf{i}'$ *is in* $\mathcal{X}$ *and is adapted to* $s_i Q$.

(b) *If* $i \to j$ *in* Q *then* α_j *precedes* α_i *in the sequence* $\mathbf{i}'$.

Proof. It is easy to see that (a) follows from the proof of Proposition 5.9. We now prove (b). Assume that α_i precedes α_j in the sequence (5.11). If we had $j = i_1$, then $\alpha_j = \beta_1$ clearly precedes α_i and we have a contradiction; thus $j \neq i_1$. Then there is an edge $i \to j$ in $s_{i_1} Q$, and the sequence (5.11) has the following form:

$$\alpha_{i_1}, \ldots, \alpha_i, \ldots, \alpha_i, \ldots$$

if $a_{ii_1} = 0$ and

$$\alpha_{i_1}, \ldots, \alpha_{i_1} + \alpha_i, \ldots, \alpha_i, \ldots, \alpha_j, \ldots$$

if $a_{ii_1} = -1$. The sequence analogous to (5.11) corresponding to $\mathbf{i}'$ above is

$$s_{i_1}(\beta_2), s_{i_1}(\beta_3), \ldots, s_{i_1}(\beta_\nu), \beta_1$$

hence is of the form

$$\cdots, \alpha_i, \ldots, \alpha_j, \ldots$$

if $a_{ii_1} = 0$ and

$$\ldots, \alpha_i, \ldots, \alpha_{i_1} + \alpha_i, \ldots, \alpha_j, \ldots$$

if $a_{ii_1} = -1$. In both cases we see that α_i precedes α_j in the sequence analogous to (5.11) corresponding to $\mathbf{i}'$. We can now repeat the previous argument for $\mathbf{i}'$ instead of $\mathbf{i}$, etc. Eventually we find a sequence that begins with j and we have a contradiction. □

From Proposition 5.9 (a) it follows that any indecomposable module in $\mathbf{Mod}(Q)$ remains indecomposable over the algebraic closure of $\Bbbk$ and that, in fact, the classification of (indecomposable) modules is independent of the ground field $\Bbbk$. It also follows that there is a bijection $\mathbf{c} \mapsto V_\mathbf{c}$ between $\mathbb{N}^\nu$ and the set of isomorphism classes of objects in $\mathbf{Mod}(Q)$, where $V_\mathbf{c}$ is the direct sum of c_k copies of $S(\beta_k)$ for $k = 1, \ldots, \nu$ (β_k are as in Proposition 5.9 (c) and $\mathbf{c} = (c_1, \ldots, c_\nu)$). We have

$$\mathbf{dim}(V_\mathbf{c}) = \sum_{k=0}^{\nu} c_k \beta_k. \tag{5.12}$$

In particular, up to isomorphism there are only finitely many modules in $\mathbf{Mod}(Q)$ of specified dimension in $\mathbb{N}^\mathbb{I}$.

Let $\mathbf{v} \in \mathbb{N}^\mathbb{I}$ and

$$E_\mathbf{v} = \bigoplus_{i \to j} \mathrm{Hom}(\Bbbk^{\mathbf{v}_i}, \Bbbk^{\mathbf{v}_j})$$

where the sum is over all arrows $i \to j$, and let $G_\mathbf{v} = \prod_i \mathrm{GL}(\mathbf{v}_i)$. The group $G_\mathbf{v}$ acts naturally on $E_\mathbf{v}$ by $(g \cdot x)_{ij} = (g_j x_{ij} g_i^{-1})$. Any point in $E_\mathbf{v}$ may be regarded as a module in $\mathbf{Mod}(Q)$ of dimension $\mathbf{v}$. Moreover, two points in $E_\mathbf{v}$ define isomorphic modules if and only if they are in the same $G_\mathbf{v}$-orbit. This gives a one-to-one correspondence between the set of $G_\mathbf{v}$-orbits on $E_\mathbf{v}$ and the set of isomorphism classes of modules of dimension $\mathbf{v}$ in $\mathbf{Mod}(Q)$. Let $\mathcal{O}_\mathbf{c}$ be the orbit corresponding to the module $V_\mathbf{c}$. Thus the $G_\mathbf{v}$-orbits on $E_\mathbf{v}$ have been indexed by the sequences $\mathbf{c} \in \mathbb{N}^\nu$ satisfying (5.12).

6 Multiplication in the Hall algebra

In this section we assume that $\Bbbk = \mathbb{F}_q$, a finite field with q elements. Let Ω be an orientation of the Dynkin graph. We fix a sequence $\mathbf{i} \in \mathscr{X}$ adapted to Ω; let $(\beta_1, \ldots, \beta_\nu)$ be the corresponding total order on R^+. Following Ringel [Rin90], we define the Hall algebra $\mathcal{H}(Q)$ to be the $\mathbb{C}$-vector space with basis $[V]$ indexed by the isomorphism classes of objects in $\mathbf{Mod}(Q)$ with the $\mathbb{C}$-algebra structure given by

$$[V'] \cdot [V''] = \sum_V g^V_{V',V''}[V],$$

where $g^V_{V',V''}$ is the number of submodules of V that are isomorphic to V'' and such that the corresponding quotient module is isomorpic to V'. This is an associative algebra with unit element (the 0 module). If i is a sink (resp. a source) of Q, we denote by $\mathcal{H}_i^+(Q)$ (resp. $\mathcal{H}_i^-(Q)$) the subspace of $\mathcal{H}(Q)$ spanned by the $[V]$ in $\mathbf{Mod}_i^+(Q)$ (resp. $\mathbf{Mod}_i^-(Q)$); it is easy to see that this is a subalgebra of $\mathcal{H}(Q)$. The following result is easily verified.

Proposition 6.1. *If i is a sink of Q, then $\Phi_i^+ : \mathbf{Mod}_i^+(Q) \to \mathbf{Mod}_i^-(Q)$ defines an algebra isomorphism $\mathcal{H}_i^+(Q) \cong \mathcal{H}_i^-(Q)$, with inverse defined by Φ_i^-.*

If $i \to j$ is an arrow in Q, we denote by $S(ij)$ the module defined by $V_i = \Bbbk$, $V_j = \Bbbk$, $V_k = 0$ for $k \neq i, j$, and $x_h = \mathrm{id}$ for $h : i \to j$. With the notations in Proposition 5.9 (a), we have $S(ij) = S(\alpha_i + \alpha_j)$. The following identities in $\mathcal{H}(Q)$ are easily verified

$$\begin{aligned} [S(i)][S(j)] &= [S(j)][S(i)] + [S(ij)], \\ [S(i)][S(ij)] &= q[S(ij)][S(i)], \\ [S(ij)][S(j)] &= q[S(i)][S(ij)]. \end{aligned} \tag{6.1}$$

It follows that

$$\begin{aligned} &[S(i)]^2[S(j)] - (q+1)[S(i)][S(j)][S(i)] + q[S(j)][S(i)]^2 = 0, \\ &[S(i)][S(j)]^2 - (q+1)[S(j)][S(i)][S(j)] + q[S(j)][S(i)]^2 = 0. \end{aligned} \tag{6.2}$$

For any $N \in \mathbb{N}$ and $\alpha \in R^+$, let $[\![N]\!]! := \prod_{s=1}^N (q^s - 1)/(q-1)$ and

$$[S(\alpha)]^{((N))} = \frac{1}{[\![N]\!]!}[S(\alpha)]^N \in \mathcal{H}(Q).$$

If $P \in \mathbf{Mod}(Q)$ is a direct sum of N copies of $S(\beta_k)$, then we have as in Proposition 5.8

$$P = \Phi_{i_1}^- \Phi_{i_2}^- \cdots \Phi_{i_{k-1}}^-(M)$$

where M is a direct sum of N copies of $S(i_k)$ (in the appropriate category). From the definition of $\mathcal{H}(Q)$ it follows immediately that $X = [S(i_k)]^{((N))}$. Now using Proposition 6.1 we deduce that

$$P = [S(\beta_k)]^{((N))}. \tag{6.3}$$

Lemma 6.2. *Let $\mathbf{i} = (i_1, \ldots, i_\nu) \in \mathscr{X}$ be adapted to Ω.*

(a) *The algebra $\mathcal{H}(Q)$ is generated by the elements $[S(i)]$ ($i \in \mathbb{I}$).*

(b) *Let $i = i_1$. The algebra $\mathcal{H}_i^+(Q)$ is generated by the elements $[S(j)]$ ($j \neq i$) and $[S(ji)]$ ($j \to i$).*

Proof. Let $\mathbf{c} \in \mathbb{N}^\nu$ and $V_\mathbf{c}$ be the corresponding module in $\mathbf{Mod}(Q)$. By Proposition 5.8 and the above arguments, we see that $[V_\mathbf{c}] = [P_1][P_2]\cdots[P_\nu]$ (product in $\mathcal{H}(Q)$) where $[P_k] = [S(\beta_k)]^{((c_k))}$, hence

$$[V_\mathbf{c}] = [S(\beta_1)]^{((c_1))}[S(\beta_2)]^{((c_2))} \cdots [S(\beta_\nu)]^{((c_\nu))}. \tag{6.4}$$

$\square$

We regard $\mathbb{C}$ as an A-algebra with v acting as multiplication by a fixed square root $q^{1/2}$ of q. Let U_q, U_q^+ be the $\mathbb{C}$-algebras obtained from the A-algebras U, U^+ by extension of scalars. Let $\mathbf{z} = (z_1, \ldots, z_n) \in \mathbb{Z}^n$ be such that $z_i - z_j = 1$ whenever $i \to j$ is an arrow in Ω[3]. Such $\mathbf{z}$ clearly exists.

Let U_q^0 be the subalgebra of U_q generated by $K_i^{\pm 1}$ for all i. It is isomorphic to the the group algebra of $\mathbb{Z}^n$. Let $\widetilde{\mathcal{H}}(Q) = U_q^0 \otimes_{\mathbb{C}} \mathcal{H}(Q)$. This may be regarded as an associative algebra containing U_q^0 and $\mathcal{H}(Q)$ as subalgebras, with the commutation rule

$$K_i[V]K_i^{-1} = q^{\sum_j a_{ij}\mathbf{v}_j/2}[V]$$

where $[V] \in \mathbf{Mod}(Q)$ has dimension $\mathbf{v}$. Let $U_q^{\geq 0} = U_q^0 \otimes_{\mathbb{C}} U_q^+$; this may be identified with the subalgebra of U_q generated by U_q^0 and U_q^+.

Proposition 6.3 (Ringel [Rin90]). *There is a unique algebra isomorphism $U_q^{\geq 0} \to \widetilde{\mathcal{H}}(Q)$ such that $K_i \mapsto K_i$, $K_i^{-1} \mapsto K_i^{-1}$, $E_i \mapsto K_i^{z_i}[S(i)]$.*

Proof. The formulas above define an algebra homomorphism: we must verify that the relations which define the algebra $U_q^{\geq 0}$ are respected; this follows easily from (6.2). This homomorphism is surjective by Lemma 6.2 (a). For any $\mathbf{v} \in \mathbb{N}^{\mathbb{I}}$, we define $U_{q,\mathbf{v}}^+ \subseteq U_q^+$ and $\mathcal{H}(Q)_{\mathbf{v}}$ in the same way as $U_{\mathbf{v}}^+$ was defined. These are finite-dimensional vector spaces (of the same dimension) that form a direct sum decomposition of U_q^+ and $\mathcal{H}(Q)$. Tensoring with U_q^0 we obtain direct sum decompositions of $U_q^{\geq 0}$ and $\widetilde{\mathcal{H}}(Q)$ into summands (indexed by $\mathbb{N}^{\mathbb{I}}$). It is clear that our homomorphism respects these direct sum decompositions. Its restriction to any summand is a surjective homomorphism between free modules of the same finite rank over U_q^0, hence is an isomorphism. The proposition follows. $\square$

The inverse of the isomorphism in Proposition 6.3 may be restricted to $\mathcal{H}(Q)$ and thus defines an imbedding of algebras

$$\Gamma : \mathcal{H}(Q) \hookrightarrow U_q, \quad [S(j)] \mapsto K_j^{-z_j}E_j \tag{6.5}$$

The following property of Γ follows easily from the definition:

Proposition 6.4. *If $V \in \mathbf{Mod}(Q)$ has dimension $\mathbf{v}$, then $\Gamma([V]) = K_1^{-z_1\mathbf{v}_1} \cdots K_n^{-z_n\mathbf{v}_n}u$, where $u \in U_q^+$.*

Similarly, if $i = i_1$, we have an imbedding of algebras $\Gamma_i : \mathcal{H}(s_iQ) \hookrightarrow U_q$ such that $\Gamma_i([S(j)]) = K_j^{-z'_j}E_j$ for $j \in \mathbb{I}$, where $z'_j = z_j$ for $j \neq i$ and $z'_i = z_j + 2$. Let $\rho_i : \mathbf{U} \to \mathbf{U}$ be the algebra isomorphism defined by

$$K_j \mapsto K_j, \quad E_j \mapsto v^{a_{ij}z_j - \delta_{ij}}K_i^{a_{ij}}E_j, \quad F_j \mapsto v^{-a_{ij}z_j + \delta_{ij}}F_jK_i^{-a_{ij}}.$$

The same formula (with $v = q^{1/2}$) defines an algebra isomorphism $\rho_i : U_q \to U_q$. With these notations we can state the following result.

Lemma 6.5. *For any $\zeta \in \mathcal{H}_i^-(s_iQ)$, we have $T_i^{-1}(\rho_i(\Gamma_i(\zeta))) = \Gamma(\Phi_i^-(\zeta))$.*

Proof. We set $v = q^{1/2}$. Using Lemma 6.2 (b) and (6.1) we see that the algebra $\mathcal{H}_i^-(s_iQ)$ is generated by the elements $[S(j)]$ $(j \neq i)$ and $[S(ij)]$ $(i \to j$ in $s_iQ)$. Hence we may assume that ζ is one of these generators. Assume first that $\zeta = [S(j)]$ where $a_{ij} = -1$. Then $z_i = z_j - 1$ and

$$\Gamma(\Phi_i^-(\zeta)) = \Gamma([S(ji)]) = \Gamma([S(j)][S(i)] - [S(i)][S(j)])$$

[3]Such a sequence $\mathbf{z}$ gives a height function on the Dynkin quiver Q.

$$\begin{aligned}
&= K_j^{-z_j} E_j K_i^{-z_i} E_i - K_i^{-z_i} E_i K_j^{-z_j} E_j \\
&= v^{-z_i} K_j^{-z_j} K_i^{-z_i} (E_j E_i - v^{-1} E_i E_j), \\
T_i^{-1}(\rho_i(\Gamma_i(\zeta))) &= T_i^{-1}(\rho_i(K_j^{-z_j} E_j)) = T_i^{-1}(v^{-z_i} K_j^{-z_j} K_i^{-1} E_j) \\
&= v^{-z_i} K_j^{-z_j} K_i^{-z_j} K_i (E_j E_i - v^{-1} E_i E_j) \\
&= v^{-z_i} K_j^{-z_j} K_i^{-z_i} (E_j E_i - v^{-1} E_i E_j)
\end{aligned}$$

as required. Assume next that $\zeta = [S(j)]$ where $a_{ij} = 0$, then

$$\Gamma(\Phi_i^-(\zeta)) = \Gamma([S(j)]) = K_j^{-z_j} E_j,$$
$$T_i^{-1}(\rho_i(\Gamma_i(\zeta))) = T_i^{-1}(\rho_i(K_j^{-z_j} E_j)) = T_i^{-1}(K_j^{-z_j} E_j) = K_j^{-z_j} E_j$$

as required. Finally, assume that $\zeta = [S(ij)] = [S(i)][S(j)] - [S(j)][S(i)]$ where $i \to j$ in $s_i Q$. Then $z_i = z_j - 1$ and

$$\begin{aligned}
\Gamma(\Phi_i^-(\zeta)) &= \Gamma([S(j)]) = K_j^{-z_j} E_j, \\
T_i^{-1}(\rho_i(\Gamma_i(\zeta))) &= T_i^{-1}(\rho_i(K_i^{-z_i-2} E_i K_j^{-z_j} E_j - K_j^{-z_j} E_j K_i^{-z_i-2} E_i)) \\
&= T_i^{-1}(\rho_i(K_i^{-z_i-2} K_j^{-z_j} (v^{-z_j} E_i E_j - v^{-z_i-2} E_j E_i))) \\
&= K_i K_j^{-z_j} v^{-z_j} T_i^{-1}(\rho_i(E_i E_j - v^{-1} E_j E_i)) \\
&= K_i K_j^{-z_j} v^{-z_j} v^{z_i+1} K_i^{-1} E_j = K_j^{-z_j} E_j
\end{aligned}$$

as required. The lemma is proved. □

Let $k \in [1, \nu]$. Then as above, we have an algebra homomorphism $\Gamma_k : \mathcal{H}(Q)_k \hookrightarrow U_q$ such that

$$\Gamma_k([S(j)]) = K_j^{-z_j^k} E_j \quad \text{for} \quad k \in \mathbb{I},$$

where $Q_k = s_{i_{k-1}} \cdots s_{i_2} s_{i_1} Q$ and z_j^k are defined inductively by

$$\begin{cases} z_j^1 = z_j, & \\ z_j^k = z_j^{k-1} & \text{for } k > 1, j \neq i_{k-1}, \\ z_j^k = z_j^{k-1} + 2 & \text{for } k > 1, j = i_{k-1}. \end{cases}$$

Let $l \in [1, k]$. The following identity holds in U_q:

$$\Gamma_l(\Phi_{i_l}^- \cdots \Phi_{i_{k-2}}^- \Phi_{i_{k-1}}^-([S(i_k)])) = q^{f/2} K T_{i_l}^{-1} \cdots T_{i_{k-2}}^{-1} T_{i_{k-1}}^{-1}(E_{i_k})$$

where $[S(i_k)]$ is regarded as an element of $\mathcal{H}(Q)_k$, f is an integer and K is a monomial in the $K_i^{\pm 1}$. This is proved by descending induction on l, using Lemma 6.5 repeatedly.

For $l = 1$, the last identity can be written in the form

$$\Gamma([S(\beta_k)]) = q^{f/2} K T_{i_1}^{-1} \cdots T_{i_{k-2}}^{-1} T_{i_{k-1}}^{-1}(E_{i_k}) \tag{6.6}$$

where f is an integer and K is a monomial in the $K_i^{\pm 1}$. If $V_{\mathbf{c}}$ is a module in $\mathbf{Mod}(Q)$ corresponding to $\mathbf{c} \in \mathbb{N}^\nu$ and $E_{\mathbf{i}}^{\mathbf{c}} \in U^+$ are as in (3.1), we have (using (6.6) and (6.4))

$$\Gamma([V_{\mathbf{c}}]) = q^{f_{\mathbf{c}}/2} K(\mathbf{v}) E_{\mathbf{i}}^{\mathbf{c}} \in U_q \tag{6.7}$$

where $f_{\mathbf{c}}$ is an integer depending on $\mathbf{c} \in \mathbb{N}^\nu$, $\mathbf{dim}(V) = \mathbf{v}$ and $K(\mathbf{v}) = K_1^{-z_1\mathbf{v}_1} \cdots K_n^{-z_n\mathbf{v}_n}$ (cf. Proposition 6.4).

The integers $f_{\mathbf{c}}$ will be described in § 8. Here we note only that they are independent of q and that they satisfy an identity which we now explain. Let $\mathbf{c} = (c_1, \dots, c_\nu) \in \mathbb{N}^\nu$ be of $\mathbf{i}$-homogeneity $\mathbf{v}$ and $\mathbf{c}_k \in \mathbb{N}^\nu$ be such that its k-th coordinate is c_k and its other coordinates are zero. Let $\mathbf{v}^k$ be the $\mathbf{i}$-homogeneity of $\mathbf{c}_k$. We have

$$\Gamma([V_{\mathbf{c}}]) = \Gamma([V_{\mathbf{c}_1}] \cdots [V_{\mathbf{c}^\nu}]) = \Gamma([V_{\mathbf{c}_1}]) \cdots \Gamma([V_{\mathbf{c}^\nu}]).$$

Using (6.7), this can be rewritten as

$$q^{f_{\mathbf{c}}/2} K(\mathbf{v}) E_{\mathbf{i}}^{\mathbf{c}} = q^{f_{\mathbf{c}_1}/2} K(\mathbf{v}^1) E_{\mathbf{i}}^{\mathbf{c}_1} \cdots q^{f_{\mathbf{c}^\nu}/2} K(\mathbf{v}^\nu) E_{\mathbf{i}}^{\mathbf{c}_\nu} \tag{6.8}$$

Using the commutation formulas

$$E_{\mathbf{i}}^{\mathbf{c}_l} K(\mathbf{v}^l) = q^{m_{l,k}/2} K(\mathbf{v}^k) E_{\mathbf{i}}^{\mathbf{c}_l} \quad (l < k)$$

where $m_{l,k} = \sum_i z_i \mathbf{v}_i^k c_l(\alpha_i, \beta_l)$, we move all K-factors in (6.8) to the left and then deduce that

$$f_{\mathbf{c}} = \sum_k f_{\mathbf{c}_k} + \sum_{l<k; i \in \mathbb{I}} z_i \mathbf{v}_i^k c_l(\alpha_i, \beta_l).$$

By substituting $c_l(\alpha_i, \beta_l) = (\alpha_i, \mathbf{v}^l) = \sum_j a_{ij} \mathbf{v}_j^l$, we finally obtain

$$f_{\mathbf{c}} = \sum_k f_{\mathbf{c}_k} + \sum_{l<k; i,j \in \mathbb{I}} a_{ij} z_i \mathbf{v}_i^k \mathbf{v}_j^l. \tag{6.9}$$

7 Dimension of orbits

In this section, unless otherwise specified, $\Bbbk$ is an algebraic closure of a finite field $\mathbb{F}_q$. We again fix an orientation Ω for our Dynkin graph, an $\mathbf{i} \in \mathscr{X}$ adapted to Ω and we consider the positive roots ordered accordingly. Recall that the multiplication of the Hall algebra $\mathcal{H}(Q)$ is defined by means of extensions of modules in $\mathbf{Mod}(Q)$. This operation naturally relates to the orbits of the representation space $E_{\mathbf{v}}$, which we shall now explain.

Let $\mathbf{v}, \mathbf{v}', \mathbf{v}'' \in \mathbb{N}^{\mathbb{I}}$ be such that $\mathbf{v} = \mathbf{v}' + \mathbf{v}''$. We define a diagram of varieties

$$E_{\mathbf{v}'} \times E_{\mathbf{v}''} \xleftarrow{p_1} E' \xrightarrow{p_2} E'' \xrightarrow{p_3} E_{\mathbf{v}} \tag{7.1}$$

as follows. A point of E'' is by definition a point $x \in E_{\mathbf{v}}$ together with a $\mathbf{v}''$-dimensional subspace W of $V = \bigoplus_i \Bbbk^{\mathbf{v}_i}$ such that $x_h(W_{s(h)}) \subseteq W_{t(h)}$ for any $h \in \Omega$. The map p_3 is the obvious one (forgetting the W). A point of E' is by definition a point (x, W) of E'' together with isomorphisms $\varphi_i'' : \Bbbk^{\mathbf{v}_i''} \cong W_i$, $\varphi_i : \Bbbk^{\mathbf{v}_i}/W_i \cong \Bbbk^{\mathbf{v}_i'}$ for $i \in \mathbb{I}$. To this point we attach the point x'' of $E_{\mathbf{v}''}$ and the point x' of $E_{\mathbf{v}'}$ induced by x. This defines the map p_1, and the map p_2 is the obvious one (forgetting the φ_i'', φ_i'). The group $G_{\mathbf{v}} \times G_{\mathbf{v}'} \times G_{\mathbf{v}''}$ acts naturally on each of the varieties so that the maps in (7.1) are compatible with this action. The action on $E_{\mathbf{v}'} \times E_{\mathbf{v}''}$ is the usual one on the factors $G_{\mathbf{v}'} \times G_{\mathbf{v}''}$ and $G_{\mathbf{v}}$ acts trivially. The action on $E_{\mathbf{v}}$ is the usual one on the factor $G_{\mathbf{v}}$ and the other two factors act trivially. The following result is easily verified:

Proposition 7.1. *The maps in (7.1) satisfy the following:*

(a) *p_2 is a principal $G_{\mathbf{v}'} \times G_{\mathbf{v}''}$ fibration.*

(b) *p_1 is a locally trivial fibration with smooth connected fibres of dimension*

$$\mathbf{v} \cdot \mathbf{v} - \langle \mathbf{v}', \mathbf{v}'' \rangle = \sum_i (\mathbf{v}_i')^2 + \sum_i (\mathbf{v}_i'')^2 + \sum_i \mathbf{v}_i' \mathbf{v}_i'' + \sum_{i \to j} \mathbf{v}_i' \mathbf{v}_j''.$$

(c) p_3 *is proper.*

Now let $\mathcal{O}_{\mathbf{c}'}$ be a $G_{\mathbf{v}'}$-orbit on $E_{\mathbf{v}'}$ and $\mathcal{O}_{\mathbf{c}''}$ be a $G_{\mathbf{v}''}$-orbit on $E_{\mathbf{v}''}$. Let $Z = p_2(p_1^{-1}(\mathcal{O}_{\mathbf{c}'} \times \mathcal{O}_{\mathbf{c}''}))$. From Proposition 7.1 (b), (c), we see that Z is a smooth connected variety of dimension

$$\dim(Z) = \dim(\mathcal{O}_{\mathbf{c}'}) + \dim(\mathcal{O}_{\mathbf{c}''}) + \sum_i \mathbf{v}'_i \mathbf{v}''_i + \sum_{i \to j} \mathbf{v}'_i \mathbf{v}''_j. \tag{7.2}$$

Now let $\mathcal{O}_{\mathbf{c}}$ be a $G_{\mathbf{v}}$-orbit in $E_{\mathbf{v}}$. It folows easily from the above definitions and Definition 5.3 that the following two conditions are equivalent (notation of (5.12)):

(a) $V_{\mathbf{c}} \cong V_{\mathbf{c}'} * V_{\mathbf{c}''}$;

(b) p_3 restricts to a bijection $Z \cong \mathcal{O}_{\mathbf{c}}$.

Hence, if (a) holds, we have

$$\dim(\mathcal{O}_{\mathbf{c}}) = \dim(\mathcal{O}_{\mathbf{c}'}) + \dim(\mathcal{O}_{\mathbf{c}''}) + \sum_i \mathbf{v}'_i \mathbf{v}''_i + \sum_{i \to j} \mathbf{v}'_i \mathbf{v}''_j. \tag{7.3}$$

In general, $p_3(Z)$ is a union of $G_{\mathbf{v}}$-orbits on $E_{\mathbf{v}}$. We shall write $(\mathcal{O}_{\mathbf{c}'}, \mathcal{O}_{\mathbf{c}''}) \Rightarrow \mathcal{O}_{\mathbf{c}}$ whenever $\mathcal{O}_{\mathbf{c}}$ is contained in $p_3(Z)$. In this case, we clearly have

$$\dim(\mathcal{O}_{\mathbf{c}}) \leq \dim(\mathcal{O}_{\mathbf{c}'}) + \dim(\mathcal{O}_{\mathbf{c}''}) + \sum_i \mathbf{v}'_i \mathbf{v}''_i + \sum_{i \to j} \mathbf{v}'_i \mathbf{v}''_j. \tag{7.4}$$

More generally, suppose that we are given $\mathbf{v}^k \in \mathbb{N}^{\mathbb{I}}$ and $\mathbf{c}_k \in \mathbb{N}^{\nu}$ ($k = 1, \ldots, m$) such that $\mathbf{dim}(V_{\mathbf{c}_k}) = \mathbf{v}^k$ for each k. We say that

$$(\mathcal{O}_{\mathbf{c}_1}, \mathcal{O}_{\mathbf{c}_2}, \ldots, \mathcal{O}_{\mathbf{c}_m}) \Rightarrow \mathcal{O}_{\mathbf{c}} \tag{7.5}$$

if there is a sequence $\mathbf{c} = \mathbf{c}^1, \mathbf{c}^2, \ldots, \mathbf{c}^m = \mathbf{c}_m$ of elements in $\mathbb{N}^{\nu}$ such that

$$(\mathcal{O}_{\mathbf{c}_k}, \mathcal{O}_{\mathbf{c}^{k+1}}) \Rightarrow \mathcal{O}_{\mathbf{c}_k} \quad \text{for} \quad k = 1, \ldots, m-1.$$

Lemma 7.2. *Assume that either condition (7.5) of the following condition is satisfied:*

$$V_{\mathbf{c}} \cong V_{\mathbf{c}_1} * (V_{\mathbf{c}_2} * (\cdots * V_{\mathbf{c}_m})). \tag{7.6}$$

Then we have

$$\dim(\mathcal{O}_{\mathbf{c}}) \leq \sum_{k=1}^{m} \dim(\mathcal{O}_{\mathbf{c}_k}) + \sum_{k<l;i} \mathbf{v}^k_i \mathbf{v}^l_i + \sum_{k<l;i \to j} \mathbf{v}^k_i \mathbf{v}^l_j \tag{7.7}$$

with equality if (7.6) holds.

Proof. For $m = 2$ this is just (7.3) and (7.4). The general case follows by applying repeatedly (7.3) and (7.4). □

Using the above arguments, we can now give a formula about the dimension of the orbit $\mathcal{O}_{\mathbf{c}}$, for any $\mathbf{c} \in \mathbb{N}^{\nu}$.

Proposition 7.3. *Let $m = \nu$ and assume that $\mathbf{c}_k$ has the same k-th coordinate as $\mathbf{c}$ and all its other coordinates are zero, for any $k \in [1, \nu]$. Then we have*

$$\dim(\mathcal{O}_{\mathbf{c}}) = \sum_{k;i \to j} \mathbf{v}^k_i \mathbf{v}^k_j + \sum_{k<l;i} \mathbf{v}^k_i \mathbf{v}^l_i + \sum_{k<l;i \to j} \mathbf{v}^k_i \mathbf{v}^l_j. \tag{7.8}$$

Proof. This can be deduced from Lemma 7.2. The condition (7.6) is satisfied (see Proposition 5.8 (b), note that in this case $V_{\mathbf{c}_k} = P_k$ in the notation of Proposition 5.8). Moreover, it is clear that $\dim(E_{\mathbf{c}_k}) = \sum_{i\to j} \mathbf{v}_i^k \mathbf{v}_j^k$. Hence it is enough to show that $\dim(\mathcal{O}_{\mathbf{v}_k}) = \dim(E_{\mathbf{c}_k})$. In fact, we have the following fact:

$$\text{For each } k \in [1,\nu], \text{ the orbit } \mathcal{O}_{\mathbf{c}_k} \text{ is open in } E_{\mathbf{v}_k}. \tag{7.9}$$

To prove this, let $Z^{\mathbf{c}}_{\mathbf{c}',\mathbf{c}''}$ be the intersection $Z \cap p_3^{-1}(\mathcal{O}_{\mathbf{c}})$ (with the notations in the begining of this section). All the varieties considered above are naturally defined over $\mathbb{F}_q$. It follows easily from the definition that

$$\text{The number of } \mathbb{F}_q\text{-rational points of } Z^{\mathbf{c}}_{\mathbf{c}',\mathbf{c}''} \text{ is } g^{V_{\mathbf{c}}}_{V_{\mathbf{c}'},V_{\mathbf{c}''}}, \tag{7.10}$$

$$\text{If } g^{V_{\mathbf{c}}}_{V_{\mathbf{c}'},V_{\mathbf{c}''}} \neq 0, \text{ then } Z^{\mathbf{c}}_{\mathbf{c}',\mathbf{c}''} \text{ is nonempty, hence } (\mathcal{O}_{\mathbf{c}'},\mathcal{O}_{\mathbf{c}''}) \Rightarrow \mathcal{O}_{\mathbf{c}}. \tag{7.11}$$

We now take $\mathbf{c}', \mathbf{c}''$ to have all coordinates equal to 0 except the k-th, which a', a'' respectively. If $\mathbf{p} = (p_1^k, \dots, p_n^k)$ and $\beta_k = \sum_i p_i^k \alpha_i$, then $\mathbf{dim}(V_{\mathbf{c}'}) = a'\mathbf{p}$, $\mathbf{dim}(V_{\mathbf{c}''}) = a''\mathbf{p}$. From (6.3) it follows that $g^{V_{\mathbf{c}}}_{V_{\mathbf{c}'},V_{\mathbf{c}''}}$ is zero unless $\mathbf{c} = \mathbf{c}' + \mathbf{c}''$, in which case it is equal to $[\![a']\!]^{-1}[\![a'']\!]^{-1}[\![a]\!]$, a polynomial in q of degree $a'a''$. Since this holds for any q, we see from (a) that $Z^{\mathbf{c}}_{\mathbf{c}',\mathbf{c}''}$ is empty unless $\mathbf{c} = \mathbf{c}' + \mathbf{c}''$ in which case it is a variety of dimension $a'a''$. If we now take $\mathbf{c} = \mathbf{c}' + \mathbf{c}''$, we see that $\dim(Z) = \dim(\mathcal{O}_{\mathbf{c}}) + a'a''$. Using the formula (7.2), we deduce that

$$\dim(\mathcal{O}_{\mathbf{c}}) = \dim(\mathcal{O}_{\mathbf{c}'}) + \dim(\mathcal{O}_{\mathbf{c}''}) + a'a''\Big(-1 + \sum_i (p_i^k)^2 + \sum_{i\to j} p_i^k p_j^k\Big). \tag{7.12}$$

Since $(\beta_k, \beta_k) = 2$, we have $1 = \sum_i (p_i^k)^2 - \sum_{i\to j} p_i^k p_j^k$. Introducing this in (7.12), we obtain

$$\dim(\mathcal{O}_{\mathbf{c}}) = \dim(\mathcal{O}_{\mathbf{c}'}) + \dim(\mathcal{O}_{\mathbf{c}''}) + 2a'a'' \sum_{i\to j} p_i^k p_j^k. \tag{7.13}$$

We show that $\dim(\mathcal{O}_{\mathbf{c}}) = \dim(E_{\mathbf{v}})$ by on a, the k-th coordinate of $\mathbf{c}$ (recall that the other coordinates are zero). When $a = 1$, this follows from 4.16(b). We can assume that $a > 1$ and that our statement is already known when a is replaced by a smaller number. Consider the identity (7.13) with $a = a' + a''$, $a' < a$, $a'' < a$. Using the induction hypothesis, we have

$$\begin{aligned}
\dim(\mathcal{O}_{\mathbf{c}}) &= \dim(E_{\mathbf{c}'}) + \dim(E_{\mathbf{c}''}) + 2a'a'' \sum_{i\to j} p_i^k p_j^k \\
&= (a')^2 \sum_{i\to j} p_i^k p_j^k + (a'')^2 \sum_{i\to j} p_i^k p_j^k + 2a'a'' \sum_{i\to j} p_i^k p_j^k \\
&= (a' + a'')^2 \sum_{i\to j} p_i^k p_j^k = \dim(E_{\mathbf{c}}).
\end{aligned}$$

This proves the lemma, and hence Proposition 7.3. □

Corollary 7.4. *In the setup of Proposition 7.3, the codimension of the orbit $\mathcal{O}_{\mathbf{c}}$ in the vector space $E_{\mathbf{v}}$ is given by*

$$\delta(\mathbf{c}) = -\sum_{k<l;i} \mathbf{v}_i^k \mathbf{v}_i^l + \sum_{k<l;i\to j} \mathbf{v}_j^k \mathbf{v}_i^l \tag{7.14}$$

Proof. We must show that the sum of the expression (7.14) with the right-hand side of the expression (7.8) is equal to $\dim(E_{\mathbf{v}})$. This sum is

$$\sum_{k;i\to j} \mathbf{v}_i^k \mathbf{v}_j^k + \sum_{k<l;i\to j} \mathbf{v}_i^k \mathbf{v}_j^l + \sum_{k<l;i\to j} \mathbf{v}_j^k \mathbf{v}_i^l = \sum_{i\to j} \Big(\sum_k \mathbf{v}_i^k\Big)\Big(\sum_k \mathbf{v}_j^k\Big) = \dim(E_{\mathbf{v}}).$$

The proposition is proved. □

Example 7.5. In the setup of Proposition 7.3, assume that $\mathbf{c}'_k \in \mathbb{N}^\nu$ ($k \in [1,\nu]$) satisfy $\dim(V_{\mathbf{c}'_k}) = \dim(V_{\mathbf{c}_k})$ for all k and $\mathbf{c}'_k \neq \mathbf{c}_k$ for some k. Let $\mathbf{c}' \in \mathbb{N}^\nu$ be such that

$$(\mathcal{O}_{\mathbf{c}'_1}, \mathcal{O}_{\mathbf{c}'_2}, \ldots, \mathcal{O}_{\mathbf{c}'_\nu}) \Rightarrow \mathcal{O}_{\mathbf{c}'}. \tag{7.15}$$

Using Lemma 7.2, we see that

$$\dim(\mathcal{O}_{\mathbf{c}'}) \leq \sum_{k=1}^{\nu} \dim(\mathcal{O}_{\mathbf{c}'_k}) + \sum_{k<l;i} \mathbf{v}_i^k \mathbf{v}_i^l + \sum_{k<l;i\to j} \mathbf{v}_i^k \mathbf{v}_j^l,$$
$$\dim(\mathcal{O}_{\mathbf{c}}) = \sum_{k=1}^{\nu} \dim(\mathcal{O}_{\mathbf{c}_k}) + \sum_{k<l;i} \mathbf{v}_i^k \mathbf{v}_i^l + \sum_{k<l;i\to j} \mathbf{v}_i^k \mathbf{v}_j^l.$$

Substracting term by term, we obtain that

$$\dim(\mathcal{O}_{\mathbf{c}}) - \dim(\mathcal{O}_{\mathbf{c}'}) \geq \sum_{k=1}^{\nu} (\dim(\mathcal{O}_{\mathbf{c}_k}) - \dim(\mathcal{O}_{\mathbf{c}'_k})). \tag{7.16}$$

From (7.9) and from our assumption on $\mathbf{c}'_k$ we deduce that $\dim(\mathcal{O}_{\mathbf{c}_k}) - \dim(\mathcal{O}_{\mathbf{c}'_k}) \geq 0$ for all k, with strict inequality for some k. This, together with (7.16) implies

$$\dim(\mathcal{O}_{\mathbf{c}}) - \dim(\mathcal{O}_{\mathbf{c}'}) > 0. \tag{7.17}$$

As an application of the results in this section, we deduce the following implication concerning coefficients in the monimial $E_{\mathbf{i}}^{\mathbf{c}'_1} E_{\mathbf{i}}^{\mathbf{c}'_2} \cdots E_{\mathbf{i}}^{\mathbf{c}'_\nu}$:

Proposition 7.6. *If $E_{\mathbf{i}}^{\mathbf{c}'}$ appears with nonzero coefficient in the product $E_{\mathbf{i}}^{\mathbf{c}'_1} E_{\mathbf{i}}^{\mathbf{c}'_2} \cdots E_{\mathbf{i}}^{\mathbf{c}'_\nu}$ with respect to the basis $B_{\mathbf{i}}$ of U^+, then (7.15) is satisfied.*

Proof. From the hypothesis, it follows that the analogous statement is true when U^+ is replaced by U_q^+ for large enough q. Using the homorphism $\mathbf{\Gamma}$ and the identity (6.7), we deduce that $V_{\mathbf{c}'}$ appears with nonzero coefficient in the product $V_{\mathbf{c}'_1} V_{\mathbf{c}'_1} \cdots V_{\mathbf{c}'_1}$ in the algebra $\mathcal{H}(Q)$ (notations relative to $\mathbb{F}_q$). Using (7.11) repeatedly, we see that the assumption of Example 7.5 is satisfied. □

8 Proof of Theorem 4.5

In this section we give an explicit formula for the number $f_{\mathbf{c}}$ defined in (6.7), and then use it to give a proof of Theorem 4.5. As we will see the codimension of the orbit $\mathcal{O}_{\mathbf{c}}$ appears in our formula (Lemma 8.2).

We assume that $\Bbbk$ is a finite field with q elements, that Ω is a fixed orientation of the Dynkin graph and that we have fixed $\mathbf{i} \in \mathcal{X}$ adapted to Ω. By changing the numbering of the simple roots if necessary, we may assume that the following condition is satisfied:

$$\text{If } i \to j \text{ is an arrow in } Q, \text{ then } j < i. \tag{8.1}$$

Let V be a module of dimension $\mathbf{v}$ in $\mathbf{Mod}(Q)$. For each $i \in \mathbb{I}$, let V^i be the be the submodule of V whose j-component coincides with the j-component of V for any $j \in [1,i]$ and whose other components are zero; let V_i be the module in $\mathbf{Mod}(Q)$ whose i-component is the same as the i-component of V and whose other components are zero. (The maps between the components of V_i are necessarily zero.) Thus we have a canonical exact sequence

$$0 \longrightarrow V^{i-1} \longrightarrow V^i \longrightarrow V_i \longrightarrow 0$$

(where we set $V^0 = 0$) and clearly, V^i has exactly one submodule (resp. quotient module) isomorphic to V^{i-1} (resp. V_i). We have $[V_i] = [S(i)]^{((\mathbf{v}_i))}$ in $\mathcal{H}(Q)$.

From the definition of the multiplication in $\mathcal{H}(Q)$, it follows that

$$[V_n]\cdots[V_2][V_1] = [S(n)]^{((\mathbf{v}_n))}\cdots[S(2)]^{((\mathbf{v}_2))}[S(1)]^{((\mathbf{v}_1))}$$

is equal to the sum of all modules of dimension $\mathbf{v}$ (one in each isomorphism class) each appearing with coefficient one in the sum. In other words, if we set

$$e(\mathbf{v}) = [S(n)]^{((\mathbf{v}_n))}\cdots[S(2)]^{((\mathbf{v}_2))}[S(1)]^{((\mathbf{v}_1))} \tag{8.2}$$

then

$$e(\mathbf{v}) = \sum_{\mathbf{c}} V_{\mathbf{c}} \tag{8.3}$$

where $\mathbf{c}$ runs over all elements of $\mathbb{N}^\nu$ of $\mathbf{i}$-homogeneity $\mathbf{v}$.

We now consider the homomorphism Γ of (6.5), defined in terms of a sequence $\mathbf{z} \in \mathbb{Z}^n$. Recall that $\Gamma([S(i)]) = K_i^{-z_i}E_i$ for all i. We have $[\![N]\!]! = q^{N(N-1)/4}[N]!$, hence

$$\begin{aligned}\Gamma([S(i)]^{((N))}) &= q^{-N(N-1)/4}(K_i^{-z_i}E_i)^{(N)}\\ &= q^{-N(N-1)/4}q^{z_i(N-1)N/2}E_i^{(N)}\\ &= q^{N(N-1)(2z_i-1)/4}K_i^{-Nz_i}E_i^{(N)}.\end{aligned}$$

It then follows that

$$\Gamma(e(\mathbf{v})) = q^{r_{\mathbf{v}}/2}K(\mathbf{v})E_n^{(\mathbf{v}_n)}E_{n-1}^{(\mathbf{v}_{n-1})}\cdots E_1^{(\mathbf{v}_1)}$$

where $K(\mathbf{v}) = K_1^{-z_1\mathbf{v}_1}\cdots K_n^{-z_n\mathbf{v}_n}$ (see Proposition 6.4) and

$$r_{\mathbf{v}} = \frac{1}{2}\sum_i \mathbf{v}_i(\mathbf{v}_i-1)(2z_i-1) - \sum_{i\to j}\mathbf{v}_i\mathbf{v}_j z_j. \tag{8.4}$$

We now apply Γ to the identity (8.3). Using (6.7), we obtain an equality of the following form in U_q:

$$q^{r_{\mathbf{v}}/2}K(\mathbf{v})E_n^{(\mathbf{v}_n)}E_{n-1}^{(\mathbf{v}_{n-1})}\cdots E_1^{(\mathbf{v}_1)} = \sum_{\mathbf{c}} q^{f_{\mathbf{c}}}K(\mathbf{d})E_{\mathbf{i}}^{\mathbf{c}} \tag{8.5}$$

where $\mathbf{c}$ runs over all elements of $\mathbb{N}^\nu$ of $\mathbf{i}$-homogeneity $\mathbf{v}$ and the integers $f_{\mathbf{c}}$ are independent of q. We can cancel $K(\mathbf{v})$ in the identity (8.5). Moreover, since (8.5) holds for all q, we must have a corresponding identity in U:

$$E_n^{(\mathbf{v}_n)}E_{n-1}^{(\mathbf{v}_{n-1})}\cdots E_1^{(\mathbf{v}_1)} = \sum_{\mathbf{c}} q^{f_{\mathbf{c}}-r_{\mathbf{v}}}E_{\mathbf{i}}^{\mathbf{c}}. \tag{8.6}$$

Lemma 8.1. *Exactly one exponent $f_{\mathbf{c}} - r_{\mathbf{v}}$ in the sum (8.6) is equal to zero.*

Proof. We associate $\mathbf{i}' \in \mathscr{X}$ to $\mathbf{i}$ as in Remark 4.4. Assume the truth of the following statement:

(a) $E_n^{(\mathbf{v}_n)}E_{n-1}^{(\mathbf{v}_{n-1})}\cdots E_1^{(\mathbf{v}_1)}$ belongs to the basis $B_{\mathbf{i}'}$ of $\mathscr{L}$ (see Proposition 4.2 (b)).

From Proposition 4.2 (c) it follows that any element of the basis $B_{\mathbf{i}'}$ can be expressed as a linear combination of elements in the basis $B_{\mathbf{i}}$ with coefficients in $\mathbb{Z}[v^{-1}]$ and that exactly one of these coefficients has constant term 1. Applying this to the element in (a), we see that the conclusion of the lemma holds.

It remains to prove (a). Using (4.6), we see that it is enough to verify that

$$E_1^{(\mathbf{v}_1)}E_2^{(\mathbf{v}_2)}\cdots E_n^{(\mathbf{v}_n)} \in B_{\mathbf{i}'}. \tag{8.7}$$

Since the roots β_k contain each positive root, we see that there exists a sequence $l(1) < l(2) < \cdots < l(n)$ in $[1, \nu]$ and a permutation σ of $\mathbb{I}$ such that

$$\alpha_{\sigma(j)} = \beta_{l(j)} = s_{i_1} s_{i_2} \cdots s_{i_{l(j-1)}}(\alpha_{i_{l(j)}}) \tag{8.8}$$

for all $j \in \mathbb{I}$. Using Proposition 2.2, it then follows that

$$T_{i_1} T_{i_2} \cdots T_{i_{l(j-1)}}(E_{i_{l(j)}}) = E_{\sigma(j)}$$

for all $j \in \mathbb{I}$, and by the definition of $B_{\mathbf{i}}$, this implies

$$E_{\sigma(1)}^{(\mathbf{v}_{\sigma(1)})} E_{\sigma(2)}^{(\mathbf{v}_{\sigma(2)})} \cdots E_{\sigma(n)}^{(\mathbf{v}_{\sigma(n)})} \in B_{\mathbf{i}}. \tag{8.9}$$

From Corollary 5.10 (b) we see that if $i \to j$ is an arrow in Q, then $\sigma^{-1}(j) < \sigma^{-1}(i)$. Since $E_i E_j = E_j E_i$ if $i \neq j$ are not joined in the Dynkin graph, this and assumption (8.1) show that the factors in (8.9) can be rearranged so that the element in (8.9) is equal to that in (8.7) and thus (8.7) holds. The lemma is proved. □

Lemma 8.2. *For any $\mathbf{c} \in \mathbb{N}^\nu$ of $\mathbf{i}$-homogeneity $\mathbf{v}$ we have*

$$f_{\mathbf{c}} - r_{\mathbf{v}} = -\delta(\mathbf{c}) \tag{8.10}$$

where $\delta(\mathbf{c})$ is the codimension of the orbit $\mathcal{O}_{\mathbf{c}}$ in $E_{\mathbf{v}}$ (cf. Corollary 7.4).

Proof. It is enough to prove (8.10) for a $(\mathbf{c}, \mathbf{v})$ under the following inductive assumption: (8.10) holds when $(\mathbf{c}, \mathbf{v})$ is replaced by $(\mathbf{c}', \mathbf{v}')$ such that at least one of the following conditions is satisfied:

(a) Any coordinate of $\mathbf{v}'$ is $\leq$ than the corresponding coordinate of $\mathbf{v}$ and $\mathbf{v}' \neq \mathbf{v}$.

(b) We have $\mathbf{v}' = \mathbf{v}$ and the dimension of the orbit $\mathcal{O}_{\mathbf{v}'}$ is strictly smaller than that of $\mathcal{O}_{\mathbf{c}}$.

Let $\mathbf{c}_k \in \mathbb{N}^\nu$ be such that its k-th coordinate is c_k and its other coordinates are zero. Let $\mathbf{v}^k$ be the $\mathbf{i}$-homogeneity of $\mathbf{c}_k$. We can rewrite (6.9) as follows:

$$f_{\mathbf{c}} = \sum_k f_{\mathbf{c}_k} - \sum_{l<k; i \to j} \mathbf{v}_i^k \mathbf{v}_j^l z_i - \sum_{l<k; i \to j} \mathbf{v}_i^l \mathbf{v}_j^k z_j + 2 \sum_{l<k; i} \mathbf{v}_i^k \mathbf{v}_i^l z_i.$$

From (8.4), we have

$$\begin{aligned} r_{\mathbf{v}} - \sum_k r_{\mathbf{v}^k} &= \sum_i \binom{\mathbf{v}_i}{2}(2z_i - 1) - \sum_{k,i} \binom{\mathbf{v}_i^k}{2}(2z_i - 1) - \sum_{i \to j} \mathbf{v}_i \mathbf{v}_j z_j + \sum_{k; i \to j} \mathbf{v}_i^k \mathbf{v}_j^k z_j \\ &= \sum_{l<k; i} \mathbf{v}_i^l \mathbf{v}_i^k (2z_i - 1) - \sum_{l<k; i \to j} (\mathbf{v}_i^l \mathbf{v}_j^k + \mathbf{v}_i^k \mathbf{v}_j^l) z_j. \end{aligned}$$

It follows that

$$\begin{aligned} f_{\mathbf{c}} - r_{\mathbf{v}} - \sum_k (f_{\mathbf{c}_k} - r_{\mathbf{v}^k}) &= - \sum_{l<k; i \to j} \mathbf{v}_i^k \mathbf{v}_j^l (z_i - z_j) + \sum_{l<k; i} \mathbf{v}_i^k \mathbf{v}_i^l \\ &= - \sum_{l<k; i \to j} \mathbf{v}_i^k \mathbf{v}_j^l + \sum_{l<k; i} \mathbf{v}_i^k \mathbf{v}_i^l \\ &= -\delta(\mathbf{c}) = -\delta(\mathbf{c}) + \sum_k \delta(\mathbf{c}_k). \end{aligned} \tag{8.11}$$

(We have used the equality $\delta(\mathbf{c}_k) = 0$.) Now (8.11) shows that if (8.10) holds for each $\mathbf{c}_k$ then it also holds for $\mathbf{c}$. Using the induction hypothesis, we see that we may assume that $\mathbf{c} = \mathbf{c}_k$

for some k. In this case, the orbit $\mathcal{O}_{\mathbf{c}}$ is open (by (7.9)). Hence for any $\mathbf{c}' \neq \mathbf{c}$ with the same $\mathbf{i}$-homogeneity as $\mathbf{c}$, we have

$$\dim(\mathcal{O}_{\mathbf{c}'}) < \dim(\mathcal{O}_{\mathbf{c}}) \tag{8.12}$$

and the induction hypothesis applies to $\mathbf{c}'$:

$$f_{\mathbf{c}'} - r_{\mathbf{v}} = -\delta(\mathbf{c}') = -\mathrm{codim}(\mathcal{O}_{\mathbf{c}'}) < 0. \tag{8.13}$$

(We have used Corollary 7.4 and (8.12).) Hence in (8.6), all exponents of v (except possibly for the single term corresponding to our fixed $\mathbf{c}$) are < 0. Since in our case $\delta(\mathbf{c}) = 0$, we must only prove that $f_{\mathbf{c}} - r_{\mathbf{v}} = 0$. Assume that this is not the case. Then in (8.6), all exponents of v, without exception, are different from 0. This contradicts Lemma 8.1, and the lemma is proved. □

Corollary 8.3. *We have*

$$E_n^{(\mathbf{v}_n)} E_{n-1}^{(\mathbf{v}_{n-1})} \cdots E_1^{(\mathbf{v}_1)} = \sum_{\mathbf{c}} v^{-\delta(\mathbf{c})} E_{\mathbf{i}}^{\mathbf{c}}$$

where the sum is taken over all $\mathbf{c} \in \mathbb{N}^\nu$ of $\mathbf{i}$-homogeneity $\mathbf{v}$ and $\delta(\mathbf{c})$ is the codimension of the orbit $\mathcal{O}_{\mathbf{c}}$.

As before, we associate $\mathbf{c}_k$, $\mathbf{v}^k$ to $\mathbf{c} \in \mathbb{N}^\nu$ and $\mathbf{v} \in \mathbb{N}^{\mathbb{I}}$ (for $k \in [1, \nu]$). We define an element in U

$$E(\mathbf{c}) = E(\mathbf{v}^1) \cdots E(\mathbf{v}^\nu) \tag{8.14}$$

where $E(\mathbf{v}^k) = E_n^{(\mathbf{v}_n^k)} E_{n-1}^{(\mathbf{v}_n^k)} \cdots E_1^{(\mathbf{v}_1^k)}$. Since the $E_{\mathbf{i}}^{\mathbf{c}}$ form an A-basis of U, we can write

$$E(\mathbf{c}) = \sum_{\mathbf{c}'} h_{\mathbf{c}'}^{\mathbf{c}} E_{\mathbf{i}}^{\mathbf{c}'} \tag{8.15}$$

where $\mathbf{c}'$ runs over the set of elements in $\mathbb{N}^\nu$ of $\mathbf{i}$-homogeneity $\mathbf{v}$ and $h_{\mathbf{c}'}^{\mathbf{c}} \in A$. From Corollary 8.3 and the results in Example 7.5, Proposition 7.6, we see that (recall that $\delta(\mathbf{c}_k) = 0$ for all k by (7.9))

$$h_{\mathbf{c}}^{\mathbf{c}} = 1, \tag{8.16}$$
$$h_{\mathbf{c}'}^{\mathbf{c}} \neq 0, \mathbf{c}' \neq \mathbf{c} \Rightarrow \dim(\mathcal{O}_{\mathbf{c}'}) < \dim(\mathbf{O}_{\mathbf{c}}). \tag{8.17}$$

It is clear that

$$\overline{E(\mathbf{c})} = E(\mathbf{c}). \tag{8.18}$$

On the other hand, the basis $E_{\mathbf{i}}^{\mathbf{c}}$ is in general not bar-invariant, and can write

$$\overline{E_{\mathbf{i}}^{\mathbf{c}}} = \sum_{\mathbf{c}'} \omega_{\mathbf{c}'}^{\mathbf{c}} E_{\mathbf{i}}^{\mathbf{c}'} \tag{8.19}$$

where $\mathbf{c}'$ runs over the set of elements in $\mathbb{N}^\nu$ of $\mathbf{i}$-homogeneity $\mathbf{v}$ and $\omega_{\mathbf{c}'}^{\mathbf{c}}$ are uniquely determined.

Proposition 8.4. *We have $\omega_{\mathbf{c}}^{\mathbf{c}} = 1$. Moreover, $\omega_{\mathbf{c}'}^{\mathbf{c}} \neq 0$ for $\mathbf{c} \neq \mathbf{c}'$ implies* $\dim(\mathcal{O}_{\mathbf{c}'}) < \dim(\mathcal{O}_{\mathbf{c}})$.

Proof. Using (8.18) and (8.19), we see that

$$\sum_{\mathbf{c}'} \overline{h_{\mathbf{c}'}^{\mathbf{c}}} \sum_{\mathbf{c}''} \omega_{\mathbf{c}''}^{\mathbf{c}'} E_{\mathbf{i}}^{\mathbf{c}''} = \sum_{\mathbf{c}''} h_{\mathbf{c}'}^{\mathbf{c}} E_{\mathbf{i}}^{\mathbf{c}''};$$

hence

$$h_{\mathbf{c}''}^{\mathbf{c}} = \sum_{\mathbf{c}'} \overline{h_{\mathbf{c}'}^{\mathbf{c}}} \omega_{\mathbf{c}''}^{\mathbf{c}'} \tag{8.20}$$

for all $\mathbf{c}$, $\mathbf{c}''$ of $\mathbf{i}$-homogeneity $\mathbf{v}$. For such $\mathbf{c}$, $\mathbf{c}''$ we say that $\mathbf{c}'' \leq \mathbf{c}$ if either $\mathbf{c} = \mathbf{c}''$ or if $\mathbf{c} \neq \mathbf{c}''$ and $\dim(\mathcal{O}_{\mathbf{c}'}) < \dim(\mathcal{O}_{\mathbf{c}})$. From (8.16), (8.17) we see that the matrix $(h_{\mathbf{c}''}^{\mathbf{c}})$ is upper triangular (with respect to $\leq$) with diagonal entries equal to 1. From (8.20) it then follows that the same must hold for the matrix $(\omega_{\mathbf{c}''}^{\mathbf{c}})$. The proposition is proved. □

We are now in a position to prove Theorem 4.5. As we have already explained, the canonical basis is obtained from the PBW type basis $E^{\mathbf{c}}_{\mathbf{i}}$ by solving a system of semilinear equations. We can explicit this idea by the following lemma:

Lemma 8.5. *Let H be a partially ordered set such that for any $i \leq j$ in H, the segement $[i, j]$ is finite. Assume that for any $i, j \in H$ we are given elements $u^j_i \in A$ such that*

$$\sum_k u^k_i \bar{u}^j_k = \delta^j_i \quad for \quad i, j \in H, \tag{8.21}$$

$$u^i_i = 1 \quad for \quad i \in H, \tag{8.22}$$

$$u^j_i = 0 \quad for \quad i > j. \tag{8.23}$$

Then the system of equations

$$Z^j_i = \sum_{k:i\leq k\leq j} u^k_i \overline{Z}^j_k, \quad \forall i \leq j \tag{8.24}$$

with unknowns $Z^i_j \in \mathbb{Z}[v^{-1}]$ $(i \leq j)$ has a unique solution such that $Z^i_i = 1$ for all $i \in H$ and $Z^j_i \in v^{-1}\mathbb{Z}[v^{-1}]$ for $i < j$.

Proof. For $i \leq j$ in H, we denote by $d(i, j)$ the maximum length of a chain starts at i and ends at j in H. Note that $d(i, j) < \infty$ by our assumption. For any $n \geq 0$, we consider the statement $\mathcal{P}_n$ which is the assertion of the lemma restricted to elements $i \leq j$ with $d(i, j) \leq n$ (note that (8.24) is meaningful for this statement). We prove $\mathcal{P}_n$ by induction on n. The case $n = 0$ is trivial, so assume now that $n > 1$. Let $i \leq j$. If $d(i, j) < n$, then Z^j_i is defined by $\mathcal{P}_{n-1}$. If $d(i, j) = n$, we note that $q = \sum_{k:i\leq k<j} \overline{Z}^k_i u^j_k$ is defined. We show that $\bar{q} + q = 0$. Indeed, using $\mathcal{P}_{n-1}$ and (8.21), (8.22), we have

$$\begin{aligned}
\bar{q} + q &= \sum_{k:i\leq k<j} Z^k_i \bar{u}^j_k + \sum_{l:i\leq l<j} \overline{Z}^l_i u^j_l \\
&= \sum_{k,l:i\leq l\leq k<j} \overline{Z}^l_i u^k_l \bar{u}^j_k + \sum_{k,l:i\leq l<k=j} \overline{Z}^l_i u^j_l \bar{u}^j_k \\
&= \sum_{k,l:i\leq l\leq k\leq j, l<j} \overline{Z}^l_i u^k_l \bar{u}^j_k = \sum_{l:i\leq l<j} \overline{Z}^l_i \delta^j_l = 0,
\end{aligned}$$

as required. Since $q \in A$ satisfies $\bar{q} + q = 0$, there is a unique element $p \in v^{-1}\mathbb{Z}[v^{-1}]$ such that $p - \bar{p} = q$, and we set $Z^j_i = p$. Then the required properties are clearly satisfied as far as $\mathcal{P}_n$ is concerned. This proves the existence in $\mathcal{P}_n$. The previous proof also shows uniqueness. The lemma is proved. □

Proof of Theorem 4.5. We apply Lemma 8.5 in the case where $H = \chi^{-1}_{\mathbf{i}}(\mathbf{v})$ (see (4.5)) for some $\mathbf{v} \in \mathbb{N}^{\mathbb{I}}$. We regard H as a partially ordered set with the partial order $\mathbf{c}' \leq \mathbf{c}$ defined in the proof of Proposition 8.4. We take u^j_i to be $\omega^{\mathbf{c}}_{\mathbf{c}'}$. This satisfies the assumption (8.21) since $^-$ is an involution. It satisfies (8.22), (8.23) by Proposition 8.4. We see that one can solve uniquely the system of equations

$$\zeta^{\mathbf{c}}_{\mathbf{c}'} = \sum_{\mathbf{c}'':\mathbf{c}'\leq\mathbf{c}''\leq\mathbf{c}} \omega^{\mathbf{c}''}_{\mathbf{c}'} \overline{\zeta^{\mathbf{c}}_{\mathbf{c}''}} \tag{8.25}$$

with unknowns $\zeta^{\mathbf{c}}_{\mathbf{c}'} \in \mathbb{Z}[v^{-1}]$, $\mathbf{c}' \leq \mathbf{c}$ in H, so that we have $\zeta^{\mathbf{c}}_{\mathbf{c}} = 1$ and $\zeta^{\mathbf{c}}_{\mathbf{c}'} \in v^{-1}\mathbb{Z}[v^{-1}]$ for all $\mathbf{c}' < \mathbf{c}$. For $\mathbf{c} \in H$, we set

$$\beta^{\mathbf{c}} = \sum_{\mathbf{c}'} \zeta^{\mathbf{c}}_{\mathbf{c}'} E^{\mathbf{c}}_{\mathbf{i}} \tag{8.26}$$

where, in the sum we take $\mathbf{c}' \in I$, $\mathbf{c}' \leq \mathbf{c}$. Let J be the family consisting of the elements $\beta^{\mathbf{c}}$ where $\mathbf{c} \in \chi^{-1}_{\mathbf{i}}(\mathbf{v})$ for various $\mathbf{v} \in \mathbb{N}^{\mathbb{I}}$. From (8.25) it follows that each element of J is fixed by bar involution. Moreover, from (8.26) and the definition we see that J is a $\mathbb{Z}[v^{-1}]$-basis of $\mathscr{L}$ and that $\pi(\beta^{\mathbf{c}}) = \pi(E_{\mathbf{i}}^{\mathbf{c}})$ so that π applies J bijectively onto B. Using the bar invariance of J, we see that $\mathscr{L} \cap \overline{\mathscr{L}}$ is precisely the free $\mathbb{Z}$-module with basis J and that Theorem 4.5 (b) holds with $\mathbf{B} = J$. The other statements of Theorem 4.5 also follow. □

Similar to (8.14), we now define in $\mathcal{H}(Q)$ elements

$$e(\mathbf{c}) = e(\mathbf{v}^1)e(\mathbf{v}^2)\cdots e(\mathbf{v}^\nu) \tag{8.27}$$

where $e(\mathbf{v}^k)$ is as in (8.2). We have

$$e(\mathbf{c}) = \sum_{\mathbf{c}'} \tilde{h}^{\mathbf{c}}_{\mathbf{c}'}[V_{\mathbf{c}'}] \tag{8.28}$$

where $\tilde{h}^{\mathbf{c}}_{\mathbf{c}'} \in \mathbb{C}$. Since each $g^{\mathbf{c}}_{\mathbf{c}',\mathbf{c}''}$ is a polynomial with integral coefficients in q (cf. [Rin90]), it follows that, as a function of q, $\tilde{h}^{\mathbf{c}}_{\mathbf{c}'}$ is a polynomial in q with integer coefficients.

With the above notations, we have

$$\begin{aligned}\Gamma(e(\mathbf{c})) &= \Gamma(e(\mathbf{v}^1))\Gamma(e(\mathbf{v}^2))\cdots\Gamma(e(\mathbf{v}^\nu))\\ &= q^{\frac{r_{\mathbf{v}^1}}{2}}K(\mathbf{v}^1)E(\mathbf{v}^1)\cdots q^{\frac{r_{\mathbf{v}^\nu}}{2}}K(\mathbf{v}^\nu)E(\mathbf{v}^\nu)\\ &= q^{\frac{r_{\mathbf{v}}+m}{2}}K(\mathbf{v})E(\mathbf{c})\end{aligned}$$

where

$$\begin{aligned}m &= -r_{\mathbf{v}} + \sum_k r_{\mathbf{v}^k}/2 + \sum_{l<k;i,j} \mathbf{v}^l_i\mathbf{v}^k_j z_j a_{ij}\\ &= -\sum_{l<k;i}\mathbf{v}^l_i\mathbf{v}^k_i(2z_i-1) + \sum_{l<k;i\to j}(\mathbf{v}^l_i\mathbf{v}^k_j + \mathbf{v}^k_i\mathbf{v}^l_j)z_j\\ &\quad + 2\sum_{l<k;i}\mathbf{v}^l_i\mathbf{v}^k_i z_i - \sum_{l<k;i\to j}\mathbf{v}^l_i\mathbf{v}^k_j z_j - \sum_{l<k;i\to j}\mathbf{v}^l_j\mathbf{v}^k_i z_i\\ &= -\sum_{l<k;i\to j}\mathbf{v}^l_j\mathbf{v}^k_i + \sum_{l<k;i}\mathbf{v}^l_i\mathbf{v}^k_i = -\delta(\mathbf{c}).\end{aligned}$$

It then follows that

$$\Gamma(e(\mathbf{c})) = q^{\frac{r_{\mathbf{v}}-\delta(\mathbf{c})}{2}}K(\mathbf{v})E(\mathbf{c}). \tag{8.29}$$

On the other hand, we can rewrite (6.7) taking into account Lemma 8.2:

$$\Gamma([V_{\mathbf{c}}]) = q^{\frac{-\delta(\mathbf{c})+r_{\mathbf{v}}}{2}}K(\mathbf{v})E^{\mathbf{c}}_{\mathbf{i}}. \tag{8.30}$$

Applying Γ to (8.28) and compare with (8.15), we see that

$$h^{\mathbf{c}}_{\mathbf{c}'} = q^{\frac{\delta(\mathbf{c})-\delta(\mathbf{c}')}{2}}\tilde{h}^{\mathbf{c}}_{\mathbf{c}'}. \tag{8.31}$$

The left-hand side is an element of A evaluated at $v = q^{1/2}$, while to the right side is a polynomial in q with integral coefficients. Since (8.31) holds for all q, we deduce that

Proposition 8.6. *$v^{-\delta(\mathbf{c})+\delta(\mathbf{c}')}h^{\mathbf{c}}_{\mathbf{c}'} \in A$ is a polynomial in v^2 with integral coefficients.*

Recall that $\omega^{\mathbf{c}}_{\mathbf{c}'}$ and $h^{\mathbf{c}}_{\mathbf{c}'}$ are related by the equation (8.20). From Proposition 8.6, we then conclude the following corollary:

Corollary 8.7. *Let $\tilde{\omega}^{\mathbf{c}}_{\mathbf{c}'} = v^{-\delta(\mathbf{c})+\delta(\mathbf{c}')}\omega^{\mathbf{c}}_{\mathbf{c}'} \in A$. Then $\tilde{\omega}^{\mathbf{c}}_{\mathbf{c}'} \in \mathbb{Z}[v^2, v^{-2}]$.*

Proof. From (8.20), we deduce that

$$\tilde{h}^{\mathbf{c}}_{\mathbf{c}''} = \sum_{\mathbf{c}'} v^{-2\delta(\mathbf{c})+2\delta(\mathbf{c}'')}\overline{\tilde{h}^{\mathbf{c}}_{\mathbf{c}'}}\tilde{\omega}^{\mathbf{c}'}_{\mathbf{c}''}. \tag{8.32}$$

By Proposition 8.6, the matrix $(\tilde{h}^{\mathbf{c}}_{\mathbf{c}'})$ and the matrix $(v^{-2\delta(\mathbf{c})+2\delta(\mathbf{c}'')}\overline{\tilde{h}^{\mathbf{c}}_{\mathbf{c}'}})$ have entries in $\mathbb{Z}[v^2, v^{-2}]$. Moreover, these matrices are triangular with 1 on diagonal ((8.16) and (8.17)). Hence, by (8.32), the matrix $(\tilde{\omega}^{\mathbf{c}}_{\mathbf{c}'})$ must have the same properties. The proposition follows. □

9 Relation with intersection cohomology

In this section, $\Bbbk$ is an algebraic closure of a finite field $\mathbb{F}_q$. We fix an orientation Ω for our Dynkin graph Q, and we choose $\mathbf{i} \in \mathscr{X}$ adapted to Ω. Consider the partial order on $\mathbb{N}^\nu$ defined by $\mathbf{c}' \preceq \mathbf{c}''$ if $\mathbf{c}'$, $\mathbf{c}''$ have the same $\mathbf{i}$-homogeneity and $\mathcal{O}_{\mathbf{c}'}$ is contained in the Zariski closure of $\mathcal{O}_{\mathbf{c}''}$[4] (orbits as in $E_\mathbf{v}$, where $\mathbf{v} = \mathbf{dim}(V_{\mathbf{c}'}) = \mathbf{dim}(V_{\mathbf{c}''})$). It turns out that the transition coefficients of the canonical basis $\mathbf{B}$ coincides to that of the intersection cohomology of the orbits $\mathcal{O}_\mathbf{c}$, and these are upper triangular with respect to the partial order defined as above. In order to see this, the key observation is that these coefficients are solutions of the same system of semilinear equation of the form in Lemma 8.5, and we can then apply the uniqueness result.

Before going to the details, we recall that the Hall algebra can also be described by constructible function on the representation space $E_\mathbf{v}$. For this, we fix a prime number ℓ invertible in $\Bbbk$. Given $\mathbf{v} \in \mathbb{N}^\mathbb{I}$, let $\mathcal{K}_\mathbf{v}$ the vector space of $\overline{\mathbb{Q}}_\ell$-valued functions on the set of $\mathbb{F}_q$-rational points of $E_\mathbf{v}$ that are constant on the orbits of the group of $G_\mathbf{v}(\mathbb{F}_q)$ on $E_\mathbf{v}(\mathbb{F}_q)$ or, equivalently, on the set of $\mathbb{F}_q$-rational points of any $G_\mathbf{v}$-orbit on $E_\mathbf{v}$. (The isotropy groups of the $G_\mathbf{v}$ action are connected). Let $\mathbf{1}_\mathbf{c}$ be the function that equals 1 on the $\mathbb{F}_q$-rational points of $\mathcal{O}_\mathbf{c}$ and is zero elsewhere. These functions form a $\overline{\mathbb{Q}}_\ell$-basis of $\mathcal{K}_\mathbf{v}$.

Let $\mathbf{v}, \mathbf{v}', \mathbf{v}'' \in \mathbb{N}^\mathbb{I}$ be such that $\mathbf{v} = \mathbf{v}' + \mathbf{v}''$. Assume given functions $f' \in \mathcal{K}_{\mathbf{v}'}$, $f'' \in \mathcal{K}_{\mathbf{v}''}$, we can define the convolution $f = f' * f'' \in \mathcal{K}_\mathbf{v}$ as follows. Consider the diagram of varieties

$$E_{\mathbf{v}'} \times E_{\mathbf{v}''} \xleftarrow{p_1} E' \xrightarrow{p_2} E'' \xrightarrow{p_3} E_\mathbf{v}$$

as in (7.1), and set

$$f(e) = \sum_{e', e''} \frac{\#\{\hat{e} \mid p_1(\hat{e}) = (e', e''), p_2 p_3(\hat{e}) = e\}}{\#G_{\mathbf{v}'}(\mathbb{F}_q) \# G_{\mathbf{v}''}(\mathbb{F}_q)} f'(e') f''(e'')$$

where $e', e'', e, \hat{e}$ are $\mathbb{F}_q$-rational points of $E_{\mathbf{v}'}$, $E_{\mathbf{v}''}$, $E_\mathbf{v}$, E' respectively. The operation $*$ defines an associative algebra structure on $\bigoplus_\mathbf{v} \mathcal{K}_\mathbf{v}$. Let us identify the fields $\overline{\mathbb{Q}}_\ell \cong \mathbb{C}$. We have a vector space isomorphism

$$\mathcal{K} \cong \mathcal{H}(Q)$$

that takes the basis element $\mathbf{1}_\mathbf{c}$ to $V_\mathbf{c}$. Using (7.10) we see that this is an algebra isomorphism.

Let $\mathbf{A}$ be the group ring of $\overline{\mathbb{Q}}_\ell^\times$ over $\mathbb{Z}$ and $\mathbf{K}_\mathbf{v}$ be the free $\mathbf{A}$-module with basis $(\gamma_\mathbf{v})$ indexed by the $G_\mathbf{v}$-orbits $\mathcal{O}_\mathbf{c}$ on $E_\mathbf{v}$. We may identify $\mathbf{K}_\mathbf{v}$ with the Grothendieck group of the category of constructible $G_\mathbf{v}$-equivariant $\overline{\mathbb{Q}}_\ell$ sheaves on $E_\mathbf{v}$ defined over $\mathbb{F}_q$; the basis element $\gamma_\mathbf{c}$ corresponds to the constant sheaf $\overline{\mathbb{Q}}_\ell$ on the orbit $\mathcal{O}_\mathbf{c}$, extended by 0 on the complement; the Frobenius map acts as identity on the stalks at rational points of the orbit. Let $\mathcal{I}_\mathbf{c}$ be the intersection cohomology complex[5] of the closure of $\mathcal{O}_\mathbf{c}$ in $E_\mathbf{v}$ (with coefficients in $\overline{\mathbb{Q}}_\ell$) extended by zero on the complement of that closure, with the $\mathbb{F}_q$ structure such that the Frobenius map acts as identity on the stalks ($\cong \overline{\mathbb{Q}}_\ell$) of its 0-th cohomology sheaf at rational points of the orbit $\mathcal{O}_\mathbf{c}$. We associate to $\mathcal{I}_\mathbf{c}$ an element

$$\tilde{\gamma}_\mathbf{c} = \sum_{\mathbf{c}' \preceq \mathbf{c}} p^\mathbf{c}_{\mathbf{c}'} \gamma_{\mathbf{c}'} \in \mathbf{K}_\mathbf{v} \tag{9.1}$$

where $p^\mathbf{c}_{\mathbf{c}'}$ is the (formal) alternating sum of the eigenvalues (in $\overline{\mathbb{Q}}_\ell^\times$) of the Frobenius map on the stalks of the cohomology sheaves of $\mathcal{I}_\mathbf{c}$ at any $\mathbb{F}_q$ rational point of $\mathcal{O}_{\mathbf{c}'}$. (The sum is taken in $\mathbf{A}$ rather than in $\overline{\mathbb{Q}}_\ell$). We have $p^\mathbf{c}_\mathbf{c} = 1$, and we define $p^\mathbf{c}_{\mathbf{c}'} = 0$ whenever the condition $\mathbf{c}' \preceq \mathbf{c}$ is not satisfied.

[4]Warning: this partial order is different from the one used in the proof of Proposition 8.4.
[5]This differs from the IC sheaf by a dimension shift.

The convolution operation makes also sense in the context of derived categories. Consider again the diagram (7.1) with $\mathbf{v}, \mathbf{v}', \mathbf{v}'' \in \mathbb{N}^{\mathbb{I}}$ be such that $\mathbf{v} = \mathbf{v}' + \mathbf{v}''$. Assume given two orbits $\mathcal{O}_{\mathbf{c}'} \subseteq E_{\mathbf{v}}$, $\mathcal{O}_{\mathbf{c}''} \subseteq E_{\mathbf{v}''}$. Let $\mathcal{I}' = \mathcal{I}_{\mathbf{c}'}$, $\mathcal{I}'' = \mathcal{I}_{\mathbf{c}''}$ be the corresponding intersection cohomology complexes. Then $\mathcal{I}'$ (resp. $\mathcal{I}''$) is a $G_{\mathbf{v}'}$ (resp. $G_{\mathbf{v}''}$) equivariant complex; hence using Proposition 7.1, there is a well-defined complex of sheaves $\widetilde{\mathcal{I}}$ on E'' such that

$$p_2^*\widetilde{\mathcal{I}} = p_1^*(\mathcal{I}' \otimes \mathcal{I}'')$$

($\widetilde{\mathcal{I}}$ is up to shift a simple perverse sheaf on E'' defined over $\mathbb{F}_q$.) We define a complex of sheaves on $E_{\mathbf{v}}$ (defined over $\mathbb{F}_q$) by

$$\mathcal{I}' * \mathcal{I}'' = p_{3!}^*(\widetilde{\mathcal{I}}).$$

Again, by taking taking formal alternating sums of eigenvalues of Frobenius on stalks, we associate to $\mathcal{I}', \mathcal{I}'', \mathcal{I}' * \mathcal{I}''$ elements $p', p'', p' * p''$ in $\mathbf{K}_{\mathbf{v}'}, \mathbf{K}_{\mathbf{v}''}, \mathbf{K}_{\mathbf{v}}$ respectively. It is easy to see that $(p', p'') \mapsto p' * p''$ defines an $\mathbf{A}$-bilinear pairing $* : \mathbf{K}_{\mathbf{v}'} \times \mathbf{K}_{\mathbf{v}''} \to \mathbf{K}_{\mathbf{v}}$ and hence a $\mathbf{A}$-bilinear pairing $* : \mathbf{K} \times \mathbf{K} \to \mathbf{K}$ where $\mathbf{K} = \bigoplus_{\mathbf{v}} \mathbf{K}_{\mathbf{v}}$. We thus obtain an associative algebra structure on $\mathbf{K}$.

Since p_3 is proper, $\mathcal{I}' * \mathcal{I}''$ above is a pure complex. From the theory of pure complexes [BBD82, 5.4.4, 5.3.9], it follows that for any $G_{\mathbf{v}}$ orbit $\mathcal{O}_{\mathbf{c}}$ in $E_{\mathbf{v}}$ there exists a graded $\overline{\mathbb{Q}}_\ell$-vector space $\bigoplus_s L_{\mathbf{c}}^s$ with Frobenius action such that

$$p' * p'' = \tilde{\gamma}_{\mathbf{c}'}\tilde{\gamma}_{\mathbf{c}''} = \sum_{\mathbf{c}} g(\mathbf{c}, \mathbf{c}', \mathbf{c}'')\tilde{\gamma}_{\mathbf{c}} \tag{9.2}$$

where $g(\mathbf{c}, \mathbf{c}', \mathbf{c}'') \in \mathbf{A}$ is the (formal) alternating sum over s of the eigenvalues of Frobenius on $L_{\mathbf{c}}^s$ and the eigenvalues of Frobenius on $L_{\mathbf{c}}^s$ are algebraic numbers all of whose complex conjugates have absolute value $q^{s/2}$. There is a natural group homomorphism $\iota_q : \mathbf{K} \to \mathcal{K}$ that takes the basis element $\gamma_{\mathbf{c}}$ to $\mathbf{1}_{\mathbf{c}}$ and on coefficients is given by the ring homomorphism $\mathbf{A} \to \overline{\mathbb{Q}}_\ell$ that is the identity on $\overline{\mathbb{Q}}_\ell^\times$. It follows easily from the definitions that ι_q is a ring homomorphism.

Let $U_{\mathbf{A}}, U_{\mathbf{A}}^+$ be the $\mathbf{A}$-algebras obtained from U, U^+ by tensoring with $\mathbf{A}$ over A, where $\mathbf{A}$ is regarded as a A-algebra via the imbedding $A \subseteq \mathbf{A}$ that takes v to $q^{1/2} \in \overline{\mathbb{Q}}_\ell^\times$. Consider the ring homomorphism $^- : \mathbf{A} \to \mathbf{A}$ that takes each element in $\overline{\mathbb{Q}}_\ell^\times$ to its inverse. The imbedding $A \subseteq \mathbf{A}$ considered above is compatible with the bar involutions. Hence the bar involutions extend to ring involutions of $U_{\mathbf{A}}$ and $U_{\mathbf{A}}^+$ (on the other hand, the bar involution is not well defined on U_q and U_q^+).

Proposition 9.1. *For $j \in \mathbb{I}$, let γ_j be the unique element $\gamma_{\mathbf{c}}$ in the standard basis of $\mathbf{K}_{\mathbf{v}}$ where $\mathbf{v} = \alpha_j$.*

(a) *The elements γ_j $(1 \le j \le n)$ generate $\mathbf{K}$ as an $\mathbf{A}$-algebra.*

(b) *Let $\mathbf{z} = (z_1, \ldots, z_n) \in \mathbb{Z}^n$ be such that $z_i - z_j = 1$ whenever $i \to j$ is an arrow in Q. There is a unique imbedding of $\mathbf{A}$-algebras $\mathbf{\Gamma} : \mathbf{K} \hookrightarrow U_{\mathbf{A}}$ such that*

$$\mathbf{\Gamma}(\gamma_j) = K_j^{-z_j} E_j \quad \textit{for} \quad j \in [1, n].$$

(c) *Let*

$$s(\mathbf{v}) = \frac{1}{2}\sum_i (\mathbf{v}_i^2 - \mathbf{v}_i)(2z_i - 1) - \sum_{i \to j} \mathbf{v}_i \mathbf{v}_j z_i.$$

There is a well-defined $\mathbf{A}$-linear map $\Theta : \mathbf{K}_{\mathbf{v}} \to U_{\mathbf{A}}^+$ such that

$$\mathbf{\Gamma}(\zeta) = v^{s(\mathbf{v})} K(\mathbf{v}) \Theta(\zeta)$$

for all $\zeta \in \mathbf{K}_{\mathbf{v}}$, where $K(\mathbf{v}) = K_1^{-z_1\mathbf{v}_1} \cdots K_n^{-z_n\mathbf{v}_n}$.

(d) *If* $\mathbf{c}$ *has* $\mathbf{i}$*-homogeneity* $\mathbf{v}$*, then*

$$\Gamma(\gamma_{\mathbf{c}}) = v^{-\delta_{\mathbf{c}}+r_{\mathbf{v}}} K(\mathbf{v}) E_{\mathbf{i}}^{\mathbf{c}} \in U_{\mathbf{A}},$$

where

$$r_{\mathbf{v}} = \frac{1}{2}\sum_i (\mathbf{v}_i^2 - \mathbf{v}_i)(2z_i - 1) - \sum_{i\to j} \mathbf{v}_i \mathbf{v}_j z_j$$

and $\delta_{\mathbf{c}}$ *is the codimension of the orbit* $\mathcal{O}_{\mathbf{c}}$*.*

Proof. The analogous results for U_q and $\mathcal{H}(Q)$ are contained in Lemma 6.2 (a), Proposition 6.3 and (8.30). The proposition then follows as we can pass to powers of q and the corresponding algebras therein. □

The Verdier duality of derived category of constructible complexes on $E_{\mathbf{v}}$ induces a homomorphism $D : \mathbf{K}_{\mathbf{v}} \to \mathbf{K}_{\mathbf{v}}$ that is antilinear with respect to the bar involution of $\mathbf{A}$. We have[6]

$$D(\gamma_{\mathbf{c}}) = \sum_{\mathbf{c}'\preceq\mathbf{c}} r^{\mathbf{c}}_{\mathbf{c}'} \gamma_{\mathbf{c}'} \tag{9.3}$$

where $r^{\mathbf{c}}_{\mathbf{c}'} \in \mathbf{A}$ and $r^{\mathbf{c}}_{\mathbf{c}} = q^{-d(\mathbf{c})}$, with $d(\mathbf{c}) = \dim(\mathcal{O}_{\mathbf{c}})$. Moreover,

$$D(\tilde{\gamma}_{\mathbf{c}}) = q^{-d(\mathbf{c})} \tilde{\gamma}_{\mathbf{c}}. \tag{9.4}$$

From the definition and from Proposition 7.1, we see that

$$D(\zeta' * \zeta'') = q^m D(\zeta') * D(\zeta'')$$

for $\zeta' \in \mathbf{K}_{\mathbf{v}}$, $\zeta'' \in \mathbf{K}_{\mathbf{v}''}$ where $m = -\sum_i \mathbf{v}'_i\mathbf{v}''_i - \sum_{i\to j} \mathbf{v}'_i\mathbf{v}''_j$.

Proposition 9.2. *For any* $\zeta \in \mathbf{K}_{\mathbf{v}}$*, we have*

$$\Theta(D(\zeta)) = \overline{\Theta(\zeta)}. \tag{9.5}$$

Proof. When $\zeta = \gamma_j$ is one of the algebra generators in Proposition 9.1 (a), the identity (9.2) is obvious: we have $D(\zeta) = \zeta$, $\Theta(\zeta) = E_j$. It is then enough to verify the following statement for general ζ: if (9.2) holds for some $\zeta \in \mathbf{K}_{\mathbf{v}'}$, $\zeta'' \in \mathbf{K}_{\mathbf{v}''}$ such that $\zeta = \zeta' * \zeta''$, then it also holds for ζ. □

Recall from (8.19) the following identity in U^+:

$$\overline{E_{\mathbf{i}}^{\mathbf{c}}} = \sum_{\mathbf{c}'} \omega^{\mathbf{c}}_{\mathbf{c}'} E_{\mathbf{i}}^{\mathbf{c}'} \tag{9.6}$$

where $\omega^{\mathbf{c}}_{\mathbf{c}'} \in A$ are uniquely determined. Let $\beta^{\mathbf{c}} \in \mathbf{B}$ be the unique lement such that

$$\beta^{\mathbf{c}} = \sum_{\mathbf{c}'} \zeta^{\mathbf{c}}_{\mathbf{c}'} E_{\mathbf{i}}^{\mathbf{c}'} \tag{9.7}$$

(summation over all $\mathbf{c}'$ of $\mathbf{i}$-homogeneity $\mathbf{v}$) where $\zeta^{\mathbf{c}}_{\mathbf{c}} = 1$ and $\zeta^{\mathbf{c}}_{\mathbf{c}'} \in v^{-1}\mathbb{Z}[v^{-1}]$ for $\mathbf{c}' \neq \mathbf{c}$ (see Theorem 4.5 (a), $\beta^{\mathbf{c}}$ is the inverse image of $\pi(E_{\mathbf{i}}^{\mathbf{c}})$ under the isomorphism $\tilde{\pi} : \mathscr{L} \cap \overline{\mathscr{L}} \to \mathscr{L}/v^{-1}\mathscr{L}$). If we assume that

$$\omega^{\mathbf{c}}_{\mathbf{c}'} \neq 0 \Rightarrow \mathbf{c}' \preceq \mathbf{c}, \tag{9.8}$$

$$\omega^{\mathbf{c}}_{\mathbf{c}} = 1 \tag{9.9}$$

[6] Recall that D exchanges $j_!$ and j_*.

for all $\mathbf{c}$, then by an argument almost identical to the one in the proof of Theorem 4.5 we see that one can solve uniquely the system of equations

$$Z^{\mathbf{c}}_{\mathbf{c}'} = \sum_{\mathbf{c}'':\mathbf{c}'\preceq\mathbf{c}''\preceq\mathbf{c}} \omega^{\mathbf{c}''}_{\mathbf{c}'} \overline{Z^{\mathbf{c}}_{\mathbf{c}''}} \tag{9.10}$$

for all $\mathbf{c}' \preceq \mathbf{c}$ with unknowns $Z^{\mathbf{c}}_{\mathbf{c}'} \in \mathbb{Z}[v^{-1}]$ for $\mathbf{c}' \preceq \mathbf{c}$, so that $Z^{\mathbf{c}}_{\mathbf{c}} = 1$ and $Z^{\mathbf{c}}_{\mathbf{c}'} \in v^{-1}\mathbb{Z}[v^{-1}]$ for all $\mathbf{c}' \prec \mathbf{c}$. Moreover, in this case, as each $\beta^{\mathbf{c}}$ is bar invariant, we see that

$$\zeta^{\mathbf{c}}_{\mathbf{c}'} = \begin{cases} Z^{\mathbf{c}}_{\mathbf{c}'} & \text{if } \mathbf{c}' \preceq \mathbf{c}, \\ Z^{\mathbf{c}}_{\mathbf{c}'} = 0 & \text{otherwise.} \end{cases}$$

Now (9.9) certainly holds (see Proposition 8.4). The assumption (9.8) is also satisfied, as we shall see in the following theorem.

Theorem 9.3. *Let $\zeta^{\mathbf{c}}_{\mathbf{c}'}$ be defined as in (9.7).*

(a) *If $\mathcal{O}_{\mathbf{c}'} \nsubseteq \overline{\mathcal{O}}_{\mathbf{c}}$, then $\zeta^{\mathbf{c}}_{\mathbf{c}'} = 0$.*

(b) *If $\mathcal{O}_{\mathbf{c}'} \subset \overline{\mathcal{O}}_{\mathbf{c}}$, then $\zeta^{\mathbf{c}}_{\mathbf{c}'} = v^{d(\mathbf{c}')-d(\mathbf{c})} p^{\mathbf{c}}_{\mathbf{c}'}$ where $p^{\mathbf{c}}_{\mathbf{c}'}$ is as in (9.1).*

(c) *$\Theta(\tilde{\gamma}_{\mathbf{c}}) = v^{d(\mathbf{c})}\beta^{\mathbf{c}}$.*

Proof. We set $\tilde{\beta}^{\mathbf{c}} = \Theta(\tilde{\gamma}_{\mathbf{c}})$. From Proposition 9.2 and (9.4) we see that

$$\overline{\tilde{\beta}^{\mathbf{c}}} = q^{-d(\mathbf{c})}\tilde{\beta}^{\mathbf{c}}. \tag{9.11}$$

From Proposition 9.1 (c), (d), it follows that

$$\Theta(\gamma_{\mathbf{c}}) = v^{-s(\mathbf{v})-\delta_{\mathbf{c}}+r_{\mathbf{v}}} E^{\mathbf{c}}_{\mathbf{i}} \in U_{\mathbf{A}}.$$

Hence, applying Θ to (9.1), we have

$$\tilde{\beta}^{\mathbf{c}} = \sum_{\mathbf{c}'\preceq\mathbf{c}} v^{-s(\mathbf{v})-\delta_{\mathbf{c}'}+r_{\mathbf{v}}} p^{\mathbf{c}}_{\mathbf{c}'} E^{\mathbf{c}'}_{\mathbf{i}}. \tag{9.12}$$

Now note that

$$\begin{aligned} -s(\mathbf{v}) - \delta_{\mathbf{c}'} + r_{\mathbf{v}} &= -\frac{1}{2}\sum_i (\mathbf{v}_i^2 - \mathbf{v}_i)(2z_i - 1) + \sum_{i\to j} \mathbf{v}_i\mathbf{v}_j z_i \\ &\quad + \frac{1}{2}\sum_i (\mathbf{v}_i^2 - \mathbf{v}_i)(2z_i - 1) + \sum_{i\to j} \mathbf{v}_i\mathbf{v}_j z_j - \delta_{\mathbf{c}'} \\ &= \sum_{i\to j} \mathbf{v}_i\mathbf{v}_j - \delta_{\mathbf{c}'} = \dim(E_{\mathbf{v}}) - \delta_{\mathbf{c}'} = d(\mathbf{c}'). \end{aligned}$$

Thus we can rewrite (9.11) and (9.12) in the form

$$\Theta(\gamma_{\mathbf{c}}) = v^{d(\mathbf{c})} E^{\mathbf{c}}_{\mathbf{i}}, \tag{9.13}$$

$$\tilde{\beta}^{\mathbf{c}} = \sum_{\mathbf{c}'\preceq\mathbf{c}} v^{d(\mathbf{c}')} p^{\mathbf{c}}_{\mathbf{c}'} E^{\mathbf{c}'}_{\mathbf{i}}. \tag{9.14}$$

Applying bar involution to (9.14) and taking into account (9.11) and (9.6), we obtain (recall that $v = q^{1/2}$)

$$\sum_{\mathbf{c}'\preceq\mathbf{c}} v^{d(\mathbf{c}')-2d(\mathbf{c})} p^{\mathbf{c}}_{\mathbf{c}'} E^{\mathbf{c}'}_{\mathbf{i}} = q^{-d(\mathbf{c})}\tilde{\beta}^{\mathbf{c}} = \overline{\tilde{\beta}^{\mathbf{c}}} = \sum_{\mathbf{c}'\preceq\mathbf{c}} v^{-d(\mathbf{c}')} \overline{p^{\mathbf{c}}_{\mathbf{c}'}} \sum_{\mathbf{c}''} \omega^{\mathbf{c}'}_{\mathbf{c}''} E^{\mathbf{c}''}_{\mathbf{i}},$$

so that

$$v^{-d(\mathbf{c}'')-2d(\mathbf{c})}p^{\mathbf{c}}_{\mathbf{c}''}=\sum_{\mathbf{c}'}v^{-d(\mathbf{c}')}\overline{p^{\mathbf{c}}_{\mathbf{c}'}}\omega^{\mathbf{c}'}_{\mathbf{c}''}.$$

If we set $\mathbf{p}^{\mathbf{c}}_{\mathbf{c}'}=v^{d(\mathbf{c}')-d(\mathbf{c})}p^{\mathbf{c}}_{\mathbf{c}'}$, then

$$\mathbf{p}^{\mathbf{c}}_{\mathbf{c}''}=\sum_{\mathbf{c}'}\overline{\mathbf{p}^{\mathbf{c}}_{\mathbf{c}'}}\omega^{\mathbf{c}'}_{\mathbf{c}''} \tag{9.15}$$

for all $\mathbf{c}''$. Since the matrix $(\mathbf{p}^{\mathbf{c}}_{\mathbf{c}'})$ is upper triangular (with respect to $\preceq$) with 1 on diagonal, the same must be true (by (9.15)) for the matrix $(\omega^{\mathbf{c}}_{\mathbf{c}'})$. Thus, we see that (9.8), (9.9) hold so that we can write

$$\zeta^{\mathbf{c}}_{\mathbf{c}''}=\sum_{\mathbf{c}'}\overline{\zeta^{\mathbf{c}}_{\mathbf{c}'}}\omega^{\mathbf{c}'}_{\mathbf{c}''}$$

for all $\mathbf{c}''$. Substracting this from (9.15), we obtain

$$\mathbf{p}^{\mathbf{c}}_{\mathbf{c}''}-\zeta^{\mathbf{c}}_{\mathbf{c}''}=\sum_{\mathbf{c}'}(\overline{\mathbf{p}^{\mathbf{c}}_{\mathbf{c}'}}-\overline{\zeta^{\mathbf{c}}_{\mathbf{c}'}})\omega^{\mathbf{c}'}_{\mathbf{c}''} \tag{9.16}$$

for all $\mathbf{c}''$. Let $\mu^{\mathbf{c}}_{\mathbf{c}'}=\mathbf{p}^{\mathbf{c}}_{\mathbf{c}'}-\zeta^{\mathbf{c}}_{\mathbf{c}'}$. Since $\mathbf{p}^{\mathbf{c}}_{\mathbf{c}'}$, $\zeta^{\mathbf{c}}_{\mathbf{c}'}$, $\omega^{\mathbf{c}}_{\mathbf{c}'}$ vanish unless $\mathbf{c}'\preceq\mathbf{c}$ and $\omega^{\mathbf{c}}_{\mathbf{c}}=1$ (as we have mentioned), we can deduce from (9.16) that

$$\mu^{\mathbf{c}}_{\mathbf{c}''}-\overline{\mu^{\mathbf{c}}_{\mathbf{c}''}}=\sum_{\mathbf{c}':\mathbf{c}''\prec\mathbf{c}'\preceq\mathbf{c}}\overline{\mu^{\mathbf{c}}_{\mathbf{c}'}}\omega^{\mathbf{c}'}_{\mathbf{c}''} \tag{9.17}$$

for all $\mathbf{c}''\preceq\mathbf{c}$. We shall prove that

$$\mu^{\mathbf{c}}_{\mathbf{c}''}=0 \tag{9.18}$$

for all $\mathbf{c}''\preceq\mathbf{c}$ by induction on $d(\mathbf{c})-d(\mathbf{c}'')\geq 0$. If $\mathbf{c}=\mathbf{c}''$ then $\mathbf{p}^{\mathbf{c}}_{\mathbf{c}''}=1=\zeta^{\mathbf{c}}_{\mathbf{c}''}$, so $\mu^{\mathbf{c}}_{\mathbf{c}''}=0$. Assume now that $\mathbf{c}''\prec\mathbf{c}$. We may assume that $\mu^{\mathbf{c}}_{\mathbf{c}'}=0$ for all $\mathbf{c}'$ such that $\mathbf{c}''\prec\mathbf{c}'\preceq\mathbf{c}$. Then the right-hand side of (9.17) is zero and therefore its left-hand side is also zero:

$$\mu^{\mathbf{c}}_{\mathbf{c}''}=\overline{\mu^{\mathbf{c}}_{\mathbf{c}''}}. \tag{9.19}$$

By Gabber's purity theorem [BBD82, 5.3.4] and the characterization of intersection complex [BBD82, 1.4.24], $p^{\mathbf{c}}_{\mathbf{c}''}\in\mathbf{A}$ is a (formal) $\mathbb{Z}$-linear combination of elements of $\overline{\mathbb{Q}}_\ell^\times$ that are algebraic numbers all of whose complex conjugates have absolute value $\leq q^{(d(\mathbf{c})-d(\mathbf{c}'')-1)/2}$. Using this, together with $\zeta^{\mathbf{c}}_{\mathbf{c}''}\in v^{-1}\mathbb{Z}[v^{-1}]$ and the definition of $\mathbf{p}^{\mathbf{c}}_{\mathbf{c}''}$, we see that $\mu^{\mathbf{c}}_{\mathbf{c}''}\in\mathbf{A}$ is a (formal) $\mathbb{Z}$-linear combination of elements of $\overline{\mathbb{Q}}_\ell^\times$ that are algebraic numbers all of whose complex conjugates have absolute value $\leq q^{-1/2}$. As the bar involution acts by inverting these numbers, this is compatible with (9.19) only if both sides of (9.19) are zero. Thus, (9.18) is proved and the theorem follows. □

10 A purity result

We preserve the notations in §9. We also fix $\mathbf{c}\in\mathbb{N}^\nu$ of $\mathbf{i}$-homogeneity $\mathbf{v}$. For any $k\in[1,\nu]$, we denote by $\mathbf{c}_k$ the element of $\mathbb{N}^\nu$ whose coordinates are zero except for the k-coordinate that is the same as the kth coordinate of $\mathbf{c}$, and let $\mathbf{v}^k\in\mathbb{N}^{\mathbb{I}}$ be the $\mathbf{i}$-homogeneity of $\mathbf{c}_k$. We have $V_{\mathbf{c}}=\bigoplus_k V_{\mathbf{c}_k}$ compatibly with the module structures. We denote by $V_{\mathbf{c},i}$ the i-th component of $\mathbf{V}_{\mathbf{c}}$. The module structure of $V_{\mathbf{c}}$ corresponds to an element $x\in E_{\mathbf{v}}$.

Lemma 10.1. *Given two modules V,V' in* $\mathbf{Mod}(Q)$*, we denote*

$$L(V,V')=\bigoplus_i \mathrm{Hom}(V_i,V'_i).$$

Let $\mathrm{Hom}_{\Bbbk Q}(V,V')$ be the vector space consisting of all elements of $L(V,V')$ that are morphisms in $\mathbf{Mod}(Q)$.

(a) *We have*
$$\dim(\mathrm{Hom}_{\Bbbk Q}(V_{\mathbf{c}}, V_{\mathbf{c}})) = -\sum_{k\le l; i\to j} \mathbf{v}_i^k \mathbf{v}_j^l + \sum_{k\ge l; i} \mathbf{v}_i^k \mathbf{v}_i^l.$$

(b) *If $k > l$, then $\mathrm{Hom}_{\Bbbk Q}(V_{\mathbf{c}_k}, V_{\mathbf{c}_l}) = 0$. If $k \le l$, then*
$$\dim(\mathrm{Hom}_{\Bbbk Q}(V_{\mathbf{c}_k}, V_{\mathbf{c}_l})) = -\sum_{i\to j} \mathbf{v}_i^k \mathbf{v}_j^l + \sum_{i} \mathbf{v}_i^k \mathbf{v}_i^l.$$

Proof. We define a linear map
$$L(V_{\mathbf{c}}, V_{\mathbf{c}})/\mathrm{Hom}_{\Bbbk Q}(V_{\mathbf{c}}, V_{\mathbf{c}}) \to E_{\mathbf{v}} \tag{10.1}$$
as follows. To an element $\zeta \in L(V_{\mathbf{c}}, V_{\mathbf{c}})$ that restricts to $\zeta_i : V_{\mathbf{c},i} \to V_{\mathbf{c},i}$ for any i, we associate the element $(\phi_{ij}) \in E_{\mathbf{v}}$ given by $\phi_{ij} = \zeta_j x_{ij} - x_{ij}\zeta_i$ for all $i \to j$. This clearly factors through a map (10.1). It is also clear that (10.1) is injective.

Let $\mathscr{T}$ be the image of the map (10.1). It is easy to see that $x + \mathscr{T}$ is exactly the tangent space at x to the $G_{\mathbf{v}}$-orbit in $E_{\mathbf{v}}$ passing through x ((10.1) is in fact the differential of the action of $G_{\mathbf{v}}$ at x), hence $\dim(\mathscr{T})$ (or, equivalently, $\dim(\mathrm{Hom}_{\Bbbk Q}(V_{\mathbf{c}_k}, V_{\mathbf{c}_l}))$) is equal to the dimension of that orbit, which is given by Proposition 7.3.

Now (b) follows from Proposition 5.9 (c); assume that $k < l$, and let
$$\begin{aligned}
D_1 &= \dim(\mathrm{Hom}_{\Bbbk Q}(V_{\mathbf{c}_k}, V_{\mathbf{c}_l})),\\
D_2 &= \dim(\mathrm{Hom}_{\Bbbk Q}(V_{\mathbf{c}_k}, V_{\mathbf{c}_k})),\\
D_3 &= \dim(\mathrm{Hom}_{\Bbbk Q}(V_{\mathbf{c}_l}, V_{\mathbf{c}_l})),\\
D_4 &= \dim(\mathrm{Hom}_{\Bbbk Q}(V_{\mathbf{c}_k} \oplus V_{\mathbf{c}_l}, V_{\mathbf{c}_k} \oplus V_{\mathbf{c}_l})).
\end{aligned}$$

Then by (b), we have $D_4 = D_1 + D_2 + D_3$. Moreover, D_2, D_3, D_4 are given by formulas that are special cases of (a). Hence we can find $D_1 = D_4 - D_2 - D_3$ and we see that (c) holds in this case. Finally, if $k = l$, then (c) is a special case of (a). The lemma is proved. □

We have a direct sum decomposition $E_{\mathbf{v}} = E_{\mathbf{v}}^{k,l}$ indexed by $(k,l) \in [1,\nu] \times [1,\nu]$ where
$$E_{\mathbf{v}}^{k,l} = \bigoplus_{i\to j} \mathrm{Hom}(V_{\mathbf{c}_k,i}, V_{\mathbf{c}_l,i}).$$

Lemma 10.2. *$\mathscr{T}$ contains the subspace $\bigoplus_{k\le l} E_{\mathbf{v}}^{k,l}$ of $E_{\mathbf{v}}$.*

Proof. It is enough to show that for any $k \le l$, the map (10.1) restricts to an isomorphism
$$L(V_{\mathbf{c}_k}, V_{\mathbf{c}_l})/\mathrm{Hom}_{\Bbbk Q}(V_{\mathbf{c}_k}, V_{\mathbf{c}_l}) \to E_{\mathbf{v}}^{k,l} \tag{10.2}$$
The map (10.2) is certainly injective, since (10.1) is injective. Using Lemma 10.1 (b), we see that the first space in (10.2) has dimension $\sum_{i\to j} \mathbf{v}_i^k \mathbf{v}_j^l$, which is clearly equal to $\dim(E_{\mathbf{v}}^{k,l})$. Hence (10.2) must be an isomorphism and the lemma is proved. □

For each $t \in \Bbbk^\times$, we define an element $\lambda(t) \in G_{\mathbf{v}}$ by the requirement that its component in $\mathrm{GL}(V_{\mathbf{c},i})$ acts on the summand $V_{\mathbf{c}_k,i}$ as t^k times identity. Then it is clear that in the action of $G_{\mathbf{v}}$ on $E_{\mathbf{v}}$, $\lambda(t)$ acts on the summand $E_{\mathbf{v}}^{k,l}$ as t^{k-l} times identity. Note that x is contained in $\bigoplus_k E_{\mathbf{v}}^{k,k}$, hence it is fixed by $\lambda(t)$. Now $\mathscr{T}$ is stable under the one parameter group λ since it is the sum of its intersections with the various $E_{\mathbf{v}}^{k,l}$, which are stable. Hence there exists a linear subspace $\mathscr{T}^\perp$ that is a complement of $\mathscr{T}$ in $E_{\mathbf{v}}$ and is stable under the one parameter group λ. Note that in view of Lemma 10.2, the action of the one parameter group λ on $\mathscr{T}^\perp$ has all weights of negative integers.

Proposition 10.3. *Assume that x is an $\mathbb{F}_q$-rational point of the orbit $\mathcal{O}_{\mathbf{c}}$ in $E_{\mathbf{v}}$ and let $\mathcal{O}_{\mathbf{c}'}$ be an orbit whose closure contains $\mathcal{O}_{\mathbf{c}}$. Let $\mathscr{H}^i_x$ be the stalk at x of the i-th cohomology sheaf of the intersection cohomology complex of the closure of $\mathcal{O}_{\mathbf{c}'}$. Then all eigenvalues of the Frobenius map on $\mathscr{H}^i_x$ are algebraic numbers all of whose complex conjugates have absolute value $q^{i/2}$.*

Proof. The proof has some similarity to the proof of purity in [KL80]. We may assume that the result is true when x is replaced by a rational point in an orbit of dimension stricly bigger than that of $\mathcal{O}_{\mathbf{c}}$. Let $\mathcal{T}^{\perp}$ and λ be as above. We may assume that they are defined over $\mathbb{F}_q$. From the definition of $\mathcal{T}^{\perp}$ we see easily that $x+\mathcal{T}^{\perp}$ is a transversal slice in $E_{\mathbf{v}}$ to the $G_{\mathbf{v}}$-orbit of x, and as λ has negative weights on $\mathcal{T}^{\perp}$, it defines a linear contraction of that slice to x. Now any $G_{\mathbf{v}}$-orbit is stable under λ, hence the intersection of $x+\mathcal{T}^{\perp}$ with the closure of $\mathcal{O}_{\mathbf{c}'}$ is stable under λ. Consider the cohomology sheaves of the intersection complex of this intersection. The eigenvalues of Frobenius on its stalks at points other than x have a property like the one asserted in the proposition, by the induction hypothesis. We only have to show that they have that property at x. But this follows from [KL80, 4.5 (b)], where it is deduced from the hard Lefschetz theorem. ☐

Corollary 10.4. *The sheaf $\mathscr{H}^i_x$ vanishes for i odd, and we have*

$$\sum_i \dim(\mathscr{H}^{2i}_x)v^{2i} = v^{d(\mathbf{c}')-d(\mathbf{c})}\zeta^{\mathbf{c}'}_{\mathbf{c}}. \tag{10.3}$$

In particular, $v^{d(\mathbf{c}')-d(\mathbf{c})}\zeta^{\mathbf{c}'}_{\mathbf{c}}$ is a polynomial in v^2 in $\mathbb{N}$. All eigenvalues of Frobenius on $\mathscr{H}^{2i}_x$ are equal to q^i.

Proof. Let $\tilde{\zeta}^{\mathbf{c}'}_{\mathbf{c}} = v^{d(\mathbf{c}')-d(\mathbf{c})}\zeta^{\mathbf{c}'}_{\mathbf{c}}$. The equation

$$\zeta^{\mathbf{c}'}_{\mathbf{c}} = \sum_{\mathbf{c}''} \overline{\zeta^{\mathbf{c}'}_{\mathbf{c}''}}\omega^{\mathbf{c}''}_{\mathbf{c}}$$

implies that

$$\tilde{\zeta}^{\mathbf{c}'}_{\mathbf{c}} = \sum_{\mathbf{c}''} v^{2d(\mathbf{c}')-2d(\mathbf{c}'')}\overline{\tilde{\zeta}^{\mathbf{c}'}_{\mathbf{c}''}}\tilde{\omega}^{\mathbf{c}''}_{\mathbf{c}}$$

where $\omega^{\mathbf{c}''}_{\mathbf{c}}$ is in $\mathbb{Z}[v^2, v^{-2}]$ by Corollary 8.7. We want to show that

$$\tilde{\zeta}^{\mathbf{c}'}_{\mathbf{c}} \in \mathbb{Z}[v^2, v^{-2}]. \tag{10.4}$$

We may assume that this is known when $\mathbf{c}$ is replaced by any e" with $\mathbf{c} \prec \mathbf{c}'' \preceq \mathbf{c}'$. Then the previous equation shows that

$$\tilde{\zeta}^{\mathbf{c}'}_{\mathbf{c}} - v^{2d(\mathbf{c}')-2d(\mathbf{c})}\overline{\tilde{\zeta}^{\mathbf{c}'}_{\mathbf{c}}} \in \mathbb{Z}[v^2, v^{-2}]. \tag{10.5}$$

Now $\tilde{\zeta}^{\mathbf{c}'}_{\mathbf{c}}$ involves only powers of v with exponent $< d(\mathbf{c}')-d(\mathbf{c})-1$, while $v^{2d(\mathbf{c}')-2d(\mathbf{c})}\overline{\tilde{\zeta}^{\mathbf{c}'}_{\mathbf{c}}}$ involves only powers of v with exponent $> d(\mathbf{c}')-d(\mathbf{c})+1$. This, together with (10.5), implies (10.4). From (10.4) and Theorem 9.3 (b) we see that $p^{\mathbf{c}'}_{\mathbf{c}}$ is a formal $\mathbb{Z}$-linear combination (in $\mathbf{A}$) of integral powers of q. This fact, together with Proposition 10.3, clearly implies $\mathscr{H}^i_x = 0$ for i odd and that the eigenvalues of Frobenius on $\mathscr{H}^{2i}_x$ are equal to q^i; using again Theorem 9.3 (b) we see that (10.3) holds also. The corollary is proved. ☐

The previous result shows that the coefficients $p^{\mathbf{c}'}_{\mathbf{c}}$ in the identity

$$\tilde{\gamma}_{\mathbf{c}} = \sum_{\mathbf{c}'} p^{\mathbf{c}}_{\mathbf{c}'}\gamma_{\mathbf{c}'}$$

are (as elements of $\mathbf{A}$) formal $\mathbb{Z}$-linear combination of integral powers of q. On the other hand, from 5.12 we see that the structure constants of the algebra $\mathbf{K}$ with respect to the basis $(\gamma_{\mathbf{c}})$ are (as elements of $\mathbf{A}$) formal $\mathbb{Z}$-linear combinations of integral powers of q. From these two facts it follows that

Corollary 10.5. *The the structure constants $g(\mathbf{c}, \mathbf{c}', \mathbf{c}'')$ of the algebra $\mathbf{K}$ with respect to the basis $(\tilde{\gamma}_{\mathbf{c}})$ are formal $\mathbb{Z}$-linear combinations of integral powers of q.*

Now let $h(\mathbf{c}, \mathbf{c}', \mathbf{c}'') \in A$ be the structure constants of the algebra U^+ with respect to the basis $\beta^{\mathbf{c}}$. Thus

$$\beta^{\mathbf{c}'}\beta^{\mathbf{c}''} = \sum_{\mathbf{c}} h(\mathbf{c}, \mathbf{c}', \mathbf{c}'')\beta^{\mathbf{c}}. \tag{10.6}$$

Applying the homomorphism Γ to (9.2) and using Theorem 9.3 (c), we have

$$v^{d(\mathbf{c}')+d(\mathbf{c}'')+s(\mathbf{v}')+s(\mathbf{v}'')}K(\mathbf{v}')\beta^{\mathbf{c}'}K(\mathbf{v}'')\beta^{\mathbf{c}''} = \sum_{\mathbf{c}} g(\mathbf{c}, \mathbf{c}', \mathbf{c}'')v^{d(\mathbf{c})+s(\mathbf{v})}\beta^{\mathbf{c}}, \tag{10.7}$$

where $\mathbf{v}, \mathbf{v}', \mathbf{v}''$ are the $\mathbf{i}$-homogeneities of $\mathbf{c}$, $\mathbf{c}'$, $\mathbf{c}''$ respectively. Since

$$\beta^{\mathbf{c}'}K(\mathbf{v}'') = v^m K(\mathbf{v}'')\beta^{\mathbf{c}'}$$

with $m = \sum_{i,j} \mathbf{v}'_i\mathbf{v}''_j z_j a_{ij}$, we see from (10.6) and (10.7) that

$$h(\mathbf{c}, \mathbf{c}, \mathbf{c}'') = v^M g(\mathbf{c}, \mathbf{c}', \mathbf{c}'') \tag{10.8}$$

where

$$M = d(\mathbf{c}) - d(\mathbf{c}') - d(\mathbf{c}'') + s(\mathbf{v}) - s(\mathbf{v}') - s(\mathbf{v}'') - m.$$

With this notation, we have the following result:

Theorem 10.6. (a) *Any structure constant $g(\mathbf{c}, \mathbf{c}', \mathbf{c}'')$ of the algebra $\mathbf{K}$ with respect to the basis $(\tilde{\gamma}_{\mathbf{c}})$ is a formal linear combination with positive integer coefficients of integral powers of q.*

(b) *Any structure constant $h(\mathbf{c}, \mathbf{c}', \mathbf{c}'')$ of the algebra U^+ with respect to the basis $\mathbf{B}$ (consisting of the elements $\beta^{\mathbf{c}}$) is a linear combination with positive integer coefficients of powers of v with exponents of a fixed parity.*

Proof. Assertion (a) follows immediately by combining Corollary 10.5 with (9.2). Now (b) follows from (a) and (10.8). □

Corollary 10.7. *Let $b \in \mathbf{B}$, $i \in \mathbb{I}$ and $n \in \mathbb{N}$. Write*

$$E_i^{(n)}b = \sum_{b'} m_{b'}b'$$

with $m_{b'} \in A$. Then for each $b' \in \mathbf{B}$, $m_{b'}$ is a linear combination with positive integer coefficients of powers of v with exponents of a fixed parity; the same holds for right multiplication by $E_i^{(n)}$.

Proof. Indeed, we have $E_i^{(n)} \in \mathbf{B}$, so the assertion follows from Theorem 10.6. □

We have seen in Theorem 10.6 that the multplication constants of U^+ under the canonical basis canonical basis $\mathbf{B}$ are positive. In fact, the same is true for the comultiplication of U^+, but this is not clear from our construction of $\mathbf{B}$. In [Lus91], Lusztig have generalized the construction of $\mathbf{B}$ to arbitrary symmetrizable Cartan matrix, and it is shown that the canonical basis correspond to certain perverse sheaves on the representation space $E_{\mathbf{v}}$ of the associated quiver. With this approach, he is able to construct the multiplication and comultiplication on the level of categories, and this leads to various positivity properties of the canonical basis. Moreover, it can be shown that the canonical basis $\mathbf{B}$ "descents" to canonical basis of the Verma modules of $\mathbf{U}$, whence that of integrable representations of $\mathbf{U}$. One can consult [Lus10] for further details.

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